TANIM: AN INTELLIGENT AGRICULTURAL FRAMEWORK FOR CROP RECOMMENDATION AND SOIL HEALTH MONITORING

An Undergraduate Thesis

Presented to the
Faculty of Bachelor of Science in Computer Engineering
University of Science and Technology of Southern Philippines
Cagayan de Oro City

In Partial Fulfillment
of the Requirements for the Degree of
BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

CHRISTIAN JAMES E. APUYA
RAY SIMON L. BANTACULO
ALBERT JOHN DIAZ
ALEX GUARDAQUIVIL JR.
ROBERT ROY P. SALVO
October 2025

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF EQUATIONS

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Agriculture is important in the Philippines since it serves as the foundation of food security, livelihood, and rural development. Traditional soil management practices in the Philippines, such as those endorsed by the Department of Agriculture (DA), rely on generalized soil health monitoring and fertilizer recommendations using manual tools like the Minus-One Element Technique (MOET) and the Rapid Soil Test Device (RST). These approaches often fail to address field-specific variability, crop diversity, and seasonal dynamics, leading to suboptimal resource utilization, as reported by the Department of Agriculture Region 10 (2023). Historically, soil testing requires farmers to transport soil samples to nearby laboratories located at DA offices, a process that is often expensive, time-consuming, and inaccessible in remote areas, as highlighted by Pagaduan, J. L. (2025).

In relation to this, the Department of Agriculture – Regional Field Office 10 (DA–RFO 10 or Aggie 10) conducted a soil sampling drive for rice and corn production areas in the municipalities of Sugbongcogon, Binuangan, Magsaysay, and Gingoog City from March 26–28, 2025. Aggie 10 aimed to complete soil sampling and testing in priority sites across Region 10 by May 2025 to align with the wet cropping calendar. Despite these initiatives, farmers still largely depend on traditional and observational methods. The study by Samaniego, A. L. and Gallego, M. (2024) reported that only 1.03% of farmers use soil-test kits and another 1.03% utilize government laboratory services, while the remaining 97.94% rely on ocular observation or "eyeballing" of soil and crop conditions. Furthermore, crop selection decisions are often based on historical practices rather than data-driven insights, limiting farmers' ability to adapt to changing soil and climate conditions (Sarma et al., 2024). With advancements in technology such as the Internet of Things (IoT), there is significant potential to improve crop yields and reduce agricultural losses.

Soil health is a fundamental pillar of sustainable agriculture, underpinning crop productivity, food security, and environmental resilience. According to Lal, R. (2024), degraded soils characterized by nutrient deficiencies, imbalanced pH levels, or reduced organic matter lead to lower yields, increased susceptibility to pests, and economic losses for farmers. In Region 10 (Northern Mindanao, Philippines), a major agricultural hub producing rice, corn, banana, and cacao, maintaining optimal levels of nitrogen (N), phosphorus (P), potassium (K), and a soil pH range of 6.0–7.5 is essential for nutrient availability and healthy crop growth, as discussed by Morgan and Connolly (2013) and Sondhiya and Singh (2024). Data from the Department of Agriculture Region 10 (2024) indicate that acidic soils in Bukidnon restrict phosphorus uptake, while nitrogen deficiencies in rice fields can reduce yields by up to 20%. Moreover, proper crop selection based on soil condition and season, combined with precise fertilizer application, is crucial to maximizing productivity and minimizing environmental impacts such as nutrient runoff (Patil et al., 2023).

Advancements in Internet of Things (IoT) technologies and machine learning (ML) offer promising solutions for precision agriculture. IoT-based sensors, including NPK, pH, moisture, and temperature probes, enable continuous soil health monitoring, while machine learning algorithms analyze large and complex datasets to generate data-driven insights and recommendations (Kumar et al., 2024; Lavanya et al., 2024). Modern farming systems leverage these technologies to optimize crop production, improve fertilizer efficiency, and reduce waste, ultimately enhancing yield and sustainability (Elbasi et al., 2023). By integrating IoT sensor data with historical records, agri-weather information, and agricultural zoning data, intelligent systems can provide location-specific and season-aware recommendations tailored to the agricultural conditions of Region 10.

This study proposes a Soil Health Monitoring and Recommendation

System that leverages IoT sensors and machine learning techniques to provide zone-specific and season-specific recommendations for multi-cropping and fertilizer application in Region 10. Targeting key crops such as maize (corn), mungbean (mongo), peanut, soybean, squash (kalabasa), sweet potato (camote), cassava, taro (gabi), eggplant, tomato, pechay, and cabbage, the system integrates soil health parameters (NPK, pH, moisture, humidity, and temperature), agri-weather data, online agricultural resources, and DA zoning maps to predict suitable crops and recommend precise fertilizer strategies. By addressing field-specific variability and seasonal dynamics, the proposed system aims to enhance agricultural productivity, reduce resource waste, and promote sustainable farming practices. Aligned with the Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land), this research contributes to food security, environmental conservation, and sustainable agricultural development in the Philippines.

1.2 Statement of the Problem

Traditional soil management and crop selection practices in Region 10 remain largely generalized and manual, limiting their effectiveness in addressing field-specific agricultural conditions. Farmers primarily rely on standard guidelines and conventional assessment tools provided by the Department of

Agriculture, which do not sufficiently consider variations in soil properties, crop requirements, and seasonal environmental changes. This lack of precision often results in inefficient fertilizer application, suboptimal crop selection, reduced yields, increased production costs, and gradual soil degradation (Lal, R., 2023). These issues are particularly evident in Region 10, where diverse crops such as maize (corn), mungbean (mongo), peanut, soybean, squash (kalabasa), sweet potato (camote), cassava, taro (gabi), eggplant, tomato, pechay, and cabbage are cultivated under varying soil and climatic conditions.

Despite the importance of soil health information in agricultural decision-making, several problems persist that hinder the effective use of technology and data-driven approaches in the region. These problems include:

1. Inconsistent and limited availability of accurate, real-time soil health data (pH, NPK, salinity, moisture, and temperature) across different farming zones in Region 10, making it difficult to assess actual field conditions.

2. The absence of an integrated decision-support mechanism that combines soil health data with climate and zoning information, resulting in unreliable or generalized crop suitability and fertilizer recommendations.

3. Limited access to user-friendly digital platforms that can translate com-

plex agricultural data into clear, practical recommendations understandable to farmers in Region 10.

4. Uncertainty regarding the effectiveness and scalability of existing IoT-based agricultural monitoring systems in supporting sustainable farming practices across multiple farms and locations in Region 10.

1.3 Objectives of the Study

The main objective of this study is to develop an IoT-based crop recommender system integrated with machine learning for crop recommendation based on soil health parameters (pH, NPK, salinity, moisture, and temperature) and weather data for farmers, specifically the following:

1. Implement a machine learning model to predict suitable crops and recommend fertilizers for multi-cropping in Region 10 based on soil, climate, and zoning data.

2. Develop a mobile application to monitor soil health, crop predictions, fertilizer schedules, and multi-cropping planning for farmers.

3. Develop a website application to monitor soil health insights for different zoning, farmers data and information on their farming patterns, and multi-cropping information for each zoning.

4. Evaluate the IoT-based crop recommender system's accuracy and usability in agricultural settings in Region 10.

1.4 Significance of the Study

The significance of this study lies in its potential to transform agricultural practices in Region 10 by developing an integrated Internet of Things (IoT) and machine learning–based system for soil health monitoring, crop recommendation with multi-cropping, and fertilizer recommendation. Traditional soil testing methods provided by the Department of Agriculture are often underutilized by farmers due to inaccessibility and time-consuming procedures. As a result, many farmers rely on ocular observations, which may lead to poor crop selection and increased production risks (Samaniego, A. L. & Gallego, M., 2024). This research addresses critical limitations of traditional soil management practices, particularly the reliance on generalized recommendations and the lack of zone-specific insights, which frequently result in inefficient resource utilization.

By providing precise soil health data and data-driven crop and fertilizer recommendations, the proposed system enables farmers to optimize crop yields while reducing unnecessary fertilizer expenses. Furthermore, the integration of Department of Agriculture zoning maps and credible online datasets enhances land-use efficiency and supports sustainable farming practices aligned with environmental conservation goals. The study contributes to the advancement of precision agriculture by offering a scalable IoT and machine learning framework that promotes data-driven decision-making in farming. Ultimately, this research has the potential to improve agricultural productivity, environmental sustainability, and food security in Region 10, in support of Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land).

Farmers. The system will significantly assist farmers by providing accessible soil health monitoring data, data-driven crop recommendations, and optimized fertilizer usage. This enables farmers to reduce production risks, improve decision-making, and maximize crop yields while minimizing resource waste.

Department of Agriculture. The study will support the Department of Agriculture in Region 10 by offering timely soil health insights that can improve the accuracy of fertilizer distribution programs and enhance agricultural zoning strategies. The system may also serve as a tool for planning interventions and monitoring soil health trends to promote sustainable agricultural development.

Agriculturists. This research benefits agriculturists by serving as a reliable reference for assessing soil conditions and crop suitability. It enhances their ability to provide evidence-based recommendations to farmers, reducing dependence on manual and generalized assessment methods.

Future Researchers. The study will serve as a foundation for future research in smart farming and precision agriculture. It opens opportunities for further development, such as integrating remote sensing data, advanced machine learning models, and climate-resilient agricultural techniques. Future researchers may also adapt and expand the system to other regions or crop categories, making it a scalable model for broader agricultural applications.

1.5 Scope and Limitations

This study focuses on agricultural areas located in Region 10, Philippines, where selected crops—namely maize (corn), mungbean (mongo), peanut, soybean, squash (kalabasa), sweet potato (camote), cassava, taro (gabi), eggplant, tomato, pechay, and cabbage—are commonly cultivated. These crops are the primary focus of the system's crop recommendation and fertilizer optimization process. The system is designed to analyze soil health parameters including (1) nitrogen, (2) phosphorus, (3) potassium, (4) temperature, (5) soil moisture, (6) pH level, and (7) soil salinity. These parameters are combined with agri-weather data such as precipitation, temperature, wind speed, and leaf wetness obtained through a weather API, forming the basis for crop recommendation and fertilizer management.

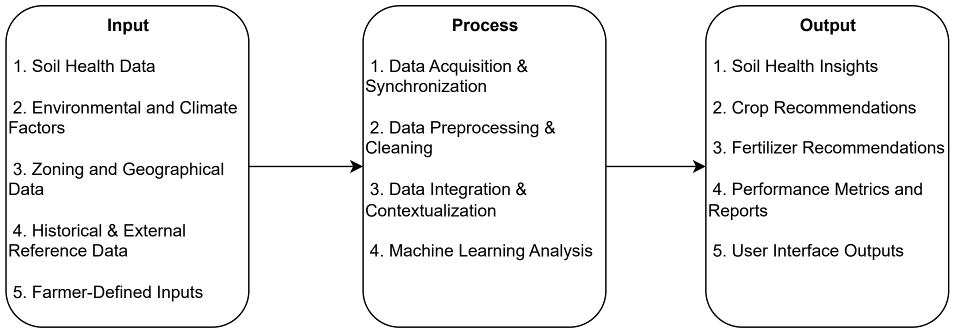
The study is geographically limited to selected farms across Bukidnon, ensuring relevance to local soil characteristics, climate conditions, and agricultural zoning data. The system emphasizes the integration of IoT-based soil sensing and machine learning techniques for data analysis and recommendation generation. However, the soil sensor used in this study is limited to seven (7) soil parameters and does not include micronutrients or soil organic matter. Additionally, real-time continuous soil monitoring is not implemented; instead, the system relies on periodic data collection and analysis. These limitations may affect the granularity of recommendations but are sufficient to demonstrate the feasibility and effectiveness of the proposed system within the defined scope.

1.5.1 Conceptual Framework

In this section, the study's conceptual framework is presented through an Input-Process-Output (IPO) model, which outlines the structure of the soil health monitoring, crop and fertilizer recommendation system. The framework integrates multiple components, each playing a vital role in developing an efficient, data-driven solution for sustainable agriculture.

Figure 1.1

IPO Model



The TANIM system follows an input–process–output (IPO) model designed to support crop and fertilizer recommendations for farmers in Region 10, Philippines. The inputs include soil health data gathered through IoT-based sensors using an ESP32 microcontroller, such as nitrogen (N), phosphorus (P), potassium (K) levels, soil pH, moisture, salinity, and temperature, along with environmental and climate data, Department of Agriculture (DA) Region˜10 soil zoning maps, historical soil information, external agricultural references, and farmer-defined parameters. These inputs undergo a systematic processing stage that involves data acquisition, preprocessing, and data cleaning to ensure reliability and consistency, followed by integration with zoning, historical, and climate datasets. The processed data are then analyzed using a LightGBM machine learning model to identify soil condition patterns, predict crop suitability, and estimate potential yield performance. Based on the analysis, the system produces outputs including soil health insights, zone-specific crop recommendations, and appropriate fertilizer suggestions, which are presented through performance metrics, reports, and a user-friendly dashboard. The performance metrics summarize model accuracy, prediction error, and data quality indicators, which are compiled into visual reports for system evaluation and validation. This data-driven approach enables farmers to make informed decisions that enhance productivity, optimize resource utilization, and promote sustainable agricultural practices.

1.5.2 Theoretical Framework

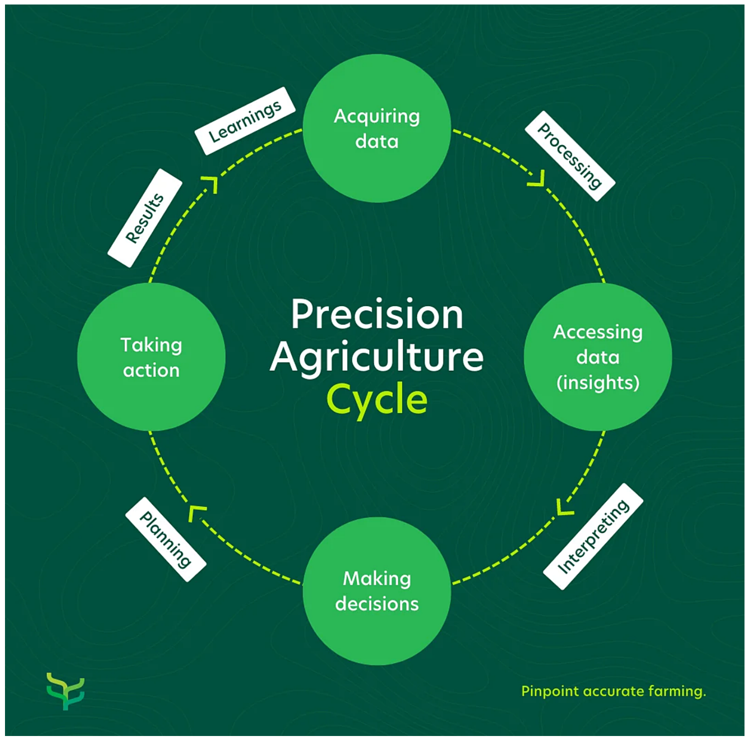
This study utilized Precision Agriculture Theory, a strategy that gathers, processes data driven insights to optimize farming practices based on conditions rather than relying on generalized farming practices recommendations, which is supported by the study of Pierre C. Robert states that an integrated farming approach designed to enhance long-term production efficiency, productivity, and profitability while minimizing environmental impacts. It involves using technology and data to optimize farming practices at a site-specific level. The theory highlights the importance of applying the right input, at the right place, at the right time, and in the right amount to maximize productivity while conserving resources.

With the integration of IoT sensors which will provide measurements of soil health parameters the precision agriculture theory will be fully operational. The soil parameters will serve as a foundation for the Multi-Criteria Decision Analysis methods to generate actionable recommendations for farmers. Moreover, machine learning boosting algorithms will be used to analyze soil and environmental patterns to predict crop suitability and yield potential.

Figure 2 explains the Precision Agriculture Cycle on how farming decisions become data-driven. It starts with acquiring data through IoT sensors and other tools that measure soil and environmental conditions. This data is then processed into insights to understand soil fertility and crop needs. Using these insights, decisions are made with the help of Machine Learning (ML) for prediction and Multi-Criteria Decision Analysis (MCDA) for evaluating multiple factors like soil parameters, climate, and zoning. The next step is taking action, where farmers apply precise fertilizers, choose suitable crops, or adjust irrigation. Finally, the results and learnings feed back into the system, ensuring continuous improvement in productivity, efficiency, and sustainability.

Figure 1.2

IPO Model



By combining ML's predictive capability with MCDA's structured decision making process, the study applies the principles of precision agriculture in a holistic manner. This ensures that recommendations for crop selection and fertilizer application are data-driven, addressing the variability of soils and seasonal conditions in Region 10. Thus, Precision Agriculture Theory serves as the guiding lens that connects IoT-based data collection, Machine Learning prediction, and MCDA supported decision-making, ultimately promoting sustainable and efficient farming practices.

1.6 Definition of Terms

This study defined the following terms that are mentioned in this study.

Agriculture The science, art, and practice of cultivating plants and livestock, serving as the foundation for food security, livelihoods, and rural development in regions like the Philippines.

Agricultural Zoning The practice of designating specific areas for agricultural use to conserve farmland, prevent urban conflicts, and promote orderly rural growth, often including minimum lot sizes and restrictions on non-farm activities.

Analytic Hierarchy Process (AHP) A structured decision-making technique that organizes complex problems into a hierarchy of goals, criteria, and alternatives, using pairwise comparisons to derive priorities and rank options.

Boosting Algorithms A family of machine learning techniques that convert weak learners into strong ones by iteratively training models to correct errors from previous ones, commonly used in regression and classification tasks.

Crops Plants or plant products grown and harvested extensively for profit or subsistence, categorized into food, feed, fiber, oil, ornamental, and industrial types based on their uses.

Fertilizer A natural or synthetic substance added to soil to supply essential nutrients like nitrogen, phosphorus, and potassium, enhancing plant growth and productivity in agriculture.

Internet of Things (IoT) in Agriculture The integration of connected sensors, devices, and networks in farming to monitor soil, weather, crops, and livestock, enabling data-driven decisions for improved efficiency and sustainability.

Machine Learning A subset of artificial intelligence that enables computers to learn from data and improve performance on tasks without explicit programming, often using algorithms to identify patterns and make predictions.

Multi-cropping The practice of growing two or more crops on the same piece of land in a single year, either sequentially or simultaneously, to enhance productivity, diversify income, and improve soil health.

NPK The abbreviation for the three primary macronutrients in soil and fertilizers: nitrogen (N), phosphorus (P), and potassium (K), essential for plant growth, development, and overall health.

Precision Agriculture A farming management strategy that uses data from sensors, GPS, and analytics to optimize inputs like water and fertilizers, responding to variability in crops for improved efficiency and sustainability.

Soil Health The continued capacity of soil to function as a vital living ecosystem that sustains plants, animals, and humans, encompassing biological, physical, and chemical properties.

Soil pH A measure of soil acidity or alkalinity on a scale from 0 to 14, indicating hydrogen ion concentration, which influences nutrient availability and plant growth.

Soil Salinity The accumulation of soluble salts in soil that can impair plant growth, often measured by electrical conductivity, with levels above 4 dS/m considered saline.

Sustainable Agriculture Farming practices that meet current food needs while preserving resources for future generations, balancing environmental health, economic viability, and social equity.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents some preliminary concepts and known results that are needed in this study.

2.1 Soil Health in Northern Mindanao

The literature underscores the critical role of soil health in sustainable agriculture and the limitations of traditional methods. Lal (2024) emphasizes soil degradation's impact on global food security, while Morgan and Connolly (2013) detail nutrient uptake mechanisms and challenges in imbalanced soils. In the Philippine context, Department of Agriculture Region X (2023) highlights regional soil issues, such as acidity in Bukidnon and nutrient shortages in rice fields, alongside generalized zoning maps. According to the study of Arlyn and Carmelito (2024), which assessed soil fertility in irrigated lowland areas across Northern Mindanao, revealing strongly to moderately acidic soils (pH 4.5-6.0) with variable organic matter content, peaking in Bukidnon at up to 3.5%. The study employed soil sampling and laboratory analysis to monitor key parameters like pH, and NPK, highlighting deficiencies in phosphorus and potassium that limit rice yields. This underscores the need for regular soil health monitoring to track degradation trends and inform site-specific interventions in Region 10's rice-dominated landscapes; this is supported by the study of **?** which uses several factors of crops and uses those inputs for random forest algorithms.

Crop production trends in Northern Mindanao, note that fertilizer inefficiencies in Bukidnon and Misamis Oriental contribute to yield gaps in corn and rice, with average applications exceeding recommendations by 15-20% based on the study of Canatoy (2018). The review recommends precision fertilizer strategies, such as variable-rate applications, to optimize nutrient use efficiency and minimize environmental impacts in Region 10's diverse agroecosystems. On the other hand, Calalang et al. (2014) reviewed soils and crop suitability in Bukidnon Highlands, highlighting how volcanic-derived soils support high-value crops like pineapple and coffee, but require lime amendments for acidity management to sustain productivity. The study links soil properties to crop performance, advocating for diversified cropping systems to enhance resilience in Northern Mindanao.

2.1.1 Zoning in Agriculture

In agriculture, zoning is the process of dividing land into management areas based on topographic features, soil fertility, and climate–all of which have a direct impact on crop suitability. Zoning has been used in several areas of the Philippines to maximize land use and improve site-specific agricultural recommendations. Watson et al. (2025) used the Food and Agriculture Organization of the United Nations (FAO) land evaluation to characterize and categorize soil series in Ilocos Norte and perform crop suitability assessments for rice, maize, garlic, onion, and tomato. According to their research, morphological characteristics and soil physicochemical characteristics are important factors in determining whether an area is considered highly suitable (S1), moderately suitable (S2), or marginally suitable (S3).

The use of zoning for particular crops is further illustrated by regional studies. In Samar Province, Poliquit et al. (2019a) evaluated crop suitability and soil fertility, identifying nutrient deficiencies like low levels of organic matter, phosphorus, and nitrogen as productivity-limiting factors. The significance of mapping local soil fertility for focused input management was highlighted by their findings. Similarly, Bato (2018) created a Geographic Information System (GIS) based banana cultivation suitability map that illustrates the spatial variations in suitability across the Philippine landscape. Slope, elevation, rainfall, and soil properties were all integrated in the study to create a visual guide that farmers and policymakers could use to identify the best places to grow bananas.

These zoning studies together make a strong case for creating smart frameworks that combine soil and crop suitability with cutting-edge technologies to give site-specific crop and fertilizer recommendations.

2.2 Machine Learning

Machine learning (ML) is becoming very useful in agriculture, it enables researchers and farmers to process huge amounts of data. It provides predictive insights related to soil health, crop recommendation, fertilizer and so on. Unlike traditional methods that rely on generalized formulas or manual observations, ML can accommodate complex and internal relationships between the soil, climate and crop parameters. This will enable them to make site-specific recommendations which will increase productivity, reduce costs and improve sustainable farming practices.

As noted by Islam et al. (2023), ML enabled systems are flexible. They not only analyze nutrient levels of soil but also suggest crops that can be grown in various environments. They do this with more accuracy and precision than manual assessments. According to Parganiha and Verma (2024), optimized ML models like LightGBM can yield predictive accuracies as high as 97% for crop yield forecasting. These models are easily scalable for larger agricultural zones like Northern Mindanao.

2.2.1 Random Forest

Random Forest (RF) is an ensemble algorithm that constructs multiple decision trees and aggregates their outputs through majority voting (classification) or averaging (regression). Incorporating bootstrapped data subsets and randomly selected features minimizes overfitting and enhances the RF model's resilience to the inaccurate data typical of many agricultural applications. This is useful in examining soils, where data quality is often compromised due to sparse sampling and other uncontrolled environmental factors.

In predicting fertilizer amounts and estimating crop inputs, **?** demonstrated its capability in capturing nonlinear relationships integrated among soil parameters and environmental factors. The study found that RF outperformed linear models in terms of accuracy, showing its adaptability across diverse farming conditions. Islam et al. (2023) integrated RF into a crop recommendation framework, successfully suggesting suitable crops based on soil nutrient levels and weather data.

2.2.2 Boosting Algorithms

With the current technologies, Islam et al. (2023) developed an ML-enabled system for soil nutrient monitoring, and crop recommendations, evaluated through field experiments to enhance productivity and sustainability.

This approach supports multi-crop applications relevant to Region 10. Parganiha and Verma (2024) proposed an optimized LightGBM model for soil analysis and crop yield prediction, incorporating data preprocessing, feature selection, and hyperparameter tuning, achieving up to 97% predictive accuracy. LightGBM's efficiency, as noted by Data Overload (2024), makes it ideal for large-scale agricultural tasks in Region 10.

Table 1: Parameters for Different Gradient Boosting Algorithms (*Salvador (2024)*)

Algorithm	**Parameter**	**Value**
Adaptive Boosting (AdaBoost)	learning_rate	0.5
	n_estimators	200
Cat Boosting (CatBoost)	depth	4
	iterations	300
	iterations	300
Gradient Boosting Machine (GBM)	learning_rate	0.3
	max_depth	3
	n_estimators	300
Light Gradient Boosting Machine (LightGBM)	learning_rate	0.1
	n_estimators	200
	num_leaves	31
Extreme Gradient Boosting (XGBoost)	learning_rate	0.3
	max_depth	3
	n_estimators	300

Table 1 outlines the hyperparameters used for training the different gradient boosting algorithms considered in this study. Each model was configured with commonly used parameter settings based on literature benchmarks and preliminary tuning to balance performance and computational efficiency. For AdaBoost, the number of estimators was set to 200 with a relatively high learning rate of 0.5 to speed up convergence. CatBoost employed a maximum tree depth of 4, 300 boosting iterations, and a learning rate of 0.3. This reflects its ability to handle categorical features effectively. The Gradient Boosting Machine was configured with a learning rate of 0.3 and a tree depth of 3, alongside 300 estimators to control complexity while maintaining accuracy. LightGBM used a lower learning rate of 0.1 with 200 estimators and 31 leaves, capitalizing on its efficiency in handling large datasets. Lastly, XGBoost was set with 300 estimators, a learning rate of 0.3, and a maximum depth of 3, representing a balanced configuration widely adopted in classification tasks.

2.2.3 Decision Tree Analysis

Decision tree analysis is a supervised machine learning approach that uses a tree-like model of decisions and their possible consequences to classify or predict outcomes. It is widely applied in agriculture because of its simplicity, interpretability, and ability to handle both categorical and numerical data. By splitting data into smaller subsets based on key attributes such as soil nutrients, moisture, pH, and climatic conditions, decision trees can provide farmers with clear and actionable recommendations.

A study of Lavanya et al. (2024) employed Gradient Boosting Decision Trees for multi-crop fertilizer recommendation, integrating soil and climate data, and reported 95% accuracy. This highlights the method's strength in managing Region 10's diverse cropping systems where multiple environmental factors interact.

Similarly, Sudha and Hooda (2022). examined the applicability of decision tree algorithms in agriculture for classification tasks, emphasizing their effectiveness in handling diverse datasets related to crop type, soil fertility, and environmental conditions. Their study concluded that decision trees not only enhance prediction accuracy but also improve the interpretability of agricultural data, enabling researchers and farmers to make evidence-based decisions. This demonstrates that decision tree-based models are reliable tools for optimizing fertilizer management, predicting yields, and supporting sustainable farming practices.

2.2.4 Multi-Criteria Decision Analysis (MCDA)

Multi-Criteria Decision Analysis (MCDA) is a decision-making tool that evaluates and compares different alternatives based on multiple, often conflicting, criteria. In agriculture and environmental management, MCDA is particularly useful for addressing complex problems where decisions must balance productivity, sustainability, cost, and environmental impact. A study of Cicciù et al. (2022) analyzed 41 papers from 1999 to 2021, revealing a surge in MCDA applications for agricultural sustainability since 2016, with France and China leading research output. The Analytic Hierarchy Process (AHP) was the most used method (11 papers), favoring compensatory approaches, while non-compensatory outranking methods like ELECTRE and PROMETHEE were less common. The Triple Bottom Line (economic, social, environmental) was applied in 68% of studies, often at the farm level, highlighting MCDA's potential to support sustainable farming systems, though its application remains underexplored, particularly for non-compensatory and hybrid methods.

MCDA integration is supported by Saaty (2008) for AHP and Kumar et al. (2024a) for IoT-ML in tropical fertilization. Yuan et al. (2022) systematically reviewed MCDA methods for rural spatial sustainability, emphasizing AHP, SAW, and TOPSIS for complex evaluations involving soil health and environmental factors. These methods enhance decision-making for crop selection and fertilizer recommendations by weighing multiple criteria like soil nutrients, climate, and zoning data.

2.3 Crop and Fertilizer Recommendation Systems

In the Philippines, crop and fertilizer recommendations remain very general and imprecise. According to Samaniego and Gallego (2024), it was found that as much as 97.94% of farmers rely on ocular observations or visual inspection of plants and soil, while only 1.03% use soil-test kits or government laboratories. This reliance on manual methods ends up either over or under-application of fertilizer, contributing to low yields and unnecessary costs. Farmers typically follow broad guidelines or personal judgement in fertilizer management, but these approaches lack the precision required to account for variations in soil health, nutrient level, and crop demands.

To address these inefficiencies, several studies have explored machine learning and data-driven approaches for fertilizer optimization. Sani et al. (2023) employed Random Forest algorithms to predict fertilizer requirements with various soil and crop inputs, achieving high accuracy. Lavanya et al. (2024) demonstrated Gradient Boosting Decision Trees (GBDT) could yield multi-crop fertilizer recommendations with an accuracy of 95% by integrating the soil and climate data. Canatoy (2018) also recommended precision fertilizers, like using different amounts of fertilizers based on location, to avoid nutrient waste and environmental impact. These findings demonstrate the potential of ML models to provide more efficient and tailored fertilizer recommendations compared to traditional methods.

Crop recommendation systems are equally important for optimizing agricultural practices. Islam et al. (2023) developed a machine learning enabled soil monitoring and crop recommendation system that uses and analyzes soil nutrient levels and environmental conditions to suggest suitable crops across multiple environments. The study highlighted the adaptability of ML in multi-cropping systems, showing that intelligent algorithms can go beyond input optimization to guide farmers in selecting crops that align with specific soil and climatic conditions.

2.4 Soil Health Sensor in Agriculture

Soil sensors play a critical role in modern agriculture as they provide accurate and real-time data about soil conditions. By measuring key parameters such as soil moisture, temperature, pH, salinity, and nutrient levels, these sensors help farmers assess overall soil health and make informed decisions for crop management. This is supported by the study of Yin et al. (2021), in which they highlighted the four aspects that will be the main focuses for future soil sensing. First is to improve the sensing performance (e.g., sensitivity and specificity) and reliability for key soil parameters, with little interference from background noise; Second is to develop a sufficient low-power consumption

Wireless Sensor Network (WSN) with powerful data processing and long-range wireless communication capability; Third is to develop versatile soil sensing platforms that can be distributed in large-scale to collect real-time soil microenvironment data continuously; and the Fourth is to develop self-powered or power-independent sensors and sensing platforms that are low-cost, reliable, and maintenance free.

The use of soil health sensors enables precision agriculture, where inputs like water, fertilizers, and pesticides can be applied more efficiently, reducing waste and minimizing environmental impact. In addition, continuous monitoring of soil health through sensors helps in early detection of soil degradation issues such as nutrient depletion, salinization, or compaction. This allows for timely interventions that improve soil productivity and sustainability. The integration of soil sensors with Internet of Things (IoT) platforms further enhances their effectiveness by enabling wireless data transmission, cloud-based analytics, and automated decision-making systems for smart farming.

Numerous studies in the field of agriculture integrate modern advancements like Internet of Things (IoT) , it is supported by the study of Sondhiya and Singh (2024), which explores Internet of Things (IoT) and Machine Learning (ML) for soil monitoring in Zea mays (corn) in Kibangay, Lantapan, Bukidnon, demonstrating improved nutrient management. The study utilized

NPK and pH sensors with an ESP32 microcontroller and LoRaWAN, achieving 95% agreement with the Department of Agriculture's Soil Test Kit. However, its single-crop focus limits applicability to Region 10's diverse crops like rice, banana, and cacao. Moreover, the study of Kumar et al. (2024a) developed an IoT-based soil monitoring system using Arduino-based NPK and moisture sensors for rice and maize, achieving 95% nutrient detection accuracy, validating the potential of IoT for monitoring.

2.5 Synthesis

The reviewed studies provide comprehensive insights into soil health, crop productivity, and machine learning applications in agriculture. Research gained in-depth understanding of the soil health challenges in Northern Mindanao, particularly with issues like soil acidity in Bukidnon and nutrient deficiencies in rice fields. Regional assessments, including those from the Department of Agriculture Region 10 (2023, 2024) and the work of Temario and Lapoot (2024), confirmed the variability of soil properties and their direct impact on yield performance. These findings are consistent with earlier insights from Canatoy (2018), who noted the inefficiencies in fertilizer applications, and from Calalang et al. (2014), who linked volcanic-derived soils to both opportunities and limitations for high-value crops. Together, these studies stress the importance of site-specific soil health monitoring and precision interventions to counteract degradation and sustain agricultural productivity. However, most of these studies remain limited to general recommendations or crop-specific assessments, leaving a gap in developing integrated, data-driven systems tailored to multiple crops in Northern Mindanao.

Zoning has also emerged as a crucial strategy for sustainable land use. Efforts by Watson et al. (2025) using FAO land evaluation in Ilocos Norte, the crop suitability mapping in Samar by Poliquit et al. (2019b), and the GIS-based banana suitability mapping by Bato (2018) demonstrated how soil fertility, topography, and climate can be translated into actionable agricultural zoning. These initiatives underscore the value of integrating spatial data with soil and crop requirements. Yet, these zoning studies are often region-specific and crop-focused, lacking the integration with modern technologies such as machine learning and IoT sensors that could make zoning more dynamic and adaptive to current farming needs.

The incorporation of machine learning adds a new dimension to these agricultural practices. Random Forest applications by **?** and Islam et al. (2023) showed how nonlinear soil and environmental data can be used to generate accurate fertilizer and crop recommendations, while boosting algorithms such as LightGBM, as explored by Parganiha and Verma (2024), achieved high predictive accuracies in crop yield forecasting. Complementary approaches like decision tree analysis, highlighted in the work of Lavanya et al. (2024) and Sudha and Hooda (2022), further support the adaptability and interpretability of ML models. Broader frameworks such as Multi-Criteria Decision Analysis, reviewed by Cicciù et al. (2022) and Yuan et al. (2022), provide systematic ways to balance productivity, environmental sustainability, and economic viability. While these studies prove the potential of ML and decision-support systems, their application in the Philippines context–particularly Northern Mindanao–remains scarce, and most models have yet to be validated in diverse cropping systems beyond rice and maize.

The technological foundation for these innovations is strengthened by the integration of soil health sensors and IoT systems. Yin et al. (2021) emphasized the need for reliable, low-power, and scalable soil sensing technologies, while recent Philippine-based studies such as those by Sondhiya and Singh (2024) in Bukidnon and Kumar et al. (2024b) in rice and maize systems demonstrated the practicality of IoT-enabled NPK and pH sensors in monitoring nutrient status. Despite their promise, these sensor applications are still largely crop-specific and small-scale, limiting their ability to provide generalized, multi-crop recommendations across Region 10's diverse agricultural systems.

The literature review establishes a strong foundation for developing a data-driven agricultural framework in Northern Mindanao. Soil health monitoring, zoning approaches, machine learning models, and IoT-based sensors converge to form comprehensive solutions. However, there remains a clear gap: existing studies often treat these components separately–soil fertility studies without advanced analytics, zoning without integration to real-time data, or ML models without validation in local contexts. This study aims to address these gaps by developing an IoT-based Soil Health Monitoring and Recommendation System that integrates real-time soil parameters (NPK, pH, salinity, moisture, and temperature), agri-weather data, and agricultural zoning maps with machine learning and Multi-Criteria Decision Analysis (MCDA). Unlike previous studies that focused on single crops or generalized recommendations, this system will deliver zone-specific and season-specific recommendations for both multi-cropping and fertilizer application in Region 10. By targeting key crops such as maize, mungbean, peanut, soybean, squash, sweet potato, cassava, taro, eggplant, tomato, pechay, and cabbage, the project seeks to optimize yields, reduce resource waste, and promote sustainable agriculture practices aligned with the region's diverse farming systems.

CHAPTER 3

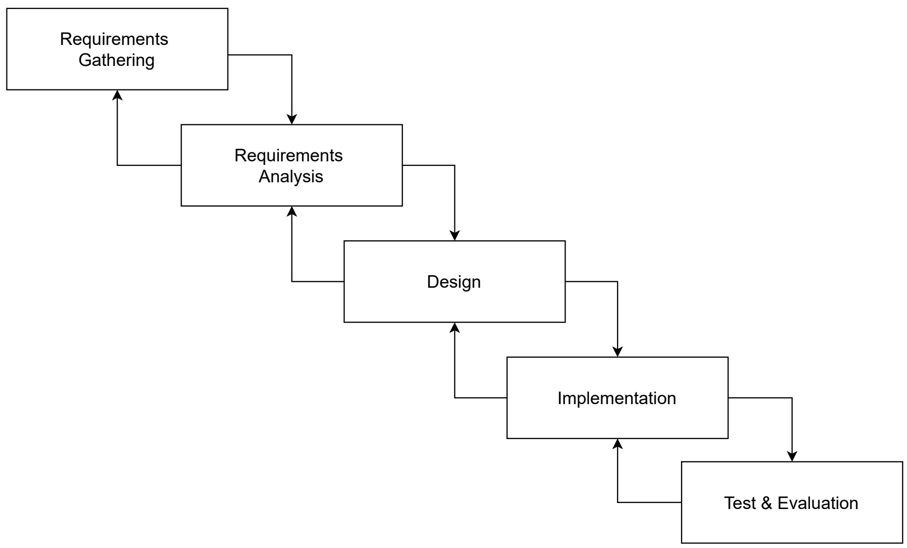
METHODOLOGY

In this chapter, the researchers outlined the methodology used to develop a soil health monitoring and recommendation system, which aimed to enhance sustainable agricultural practices through present data. The study utilized an experimental approach, integrating IoT sensors and machine learning algorithms, to enable continuous and accurate assessment of soil conditions. By focusing on essential soil parameters such as nitrogen, phosphorus, potassium , pH, salinity, moisture, and temperature this methodology provided insights that supported optimal crop management. This chapter also addressed the rationale behind the chosen experimental design, describing each stage of data collection, system setup, and analysis.

3.1 Research Design

Figure 3.1

Modified Waterfall Model



This study follows a modified waterfall research model to guide the development of the IoT-based crop and fertilizer recommendation system shown in Figure 4. The process begins with the requirements gathering phase, in which the system collects comprehensive data from the Department of Agriculture (DA) and credible online datasets. These datasets include zoning information, soil parameters, and crop-specific growth requirements. All collected datasets are integrated with the DA's zoning information to provide spatial context for agricultural analysis.

The study then proceeds to the requirements analysis phase, which identifies the need for soil health monitoring, crop prediction, and fertilizer recommendation in selected agricultural fields in Region˜10. A gradient boosting algorithm is employed using both sensor-derived soil data and zoning information as inputs. Soil health prediction focuses on classifying soil quality levels, while crop suitability and yield forecasting utilize an optimized machine learning model to estimate expected productivity based on soil and environmental conditions. Fertilizer recommendations are generated based on predicted soil nutrient deficiencies and crop requirements derived from the machine learning analysis.

The next phase is system design, where the overall system workflow is defined, providing detailed descriptions of how the IoT-based machine learning integration operates. This phase presents the system architecture, hardware and software flowcharts, and the design of the web and mobile interfaces. The design ensures seamless integration of hardware components for data acquisition, software modules for data preprocessing and prediction, and user interfaces for monitoring and decision support.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, and configuration of the required hardware and software components.

Field nodes collect and preprocess soil data before transmitting it to a central server for storage and predictive analytics. The system generates crop suitability and fertilizer recommendations, which are displayed through a web-based dashboard that visualizes soil health status, nutrient trends, weather information, farmer data, and zoning maps.

Once implementation is completed, the testing phase is conducted in the selected study area, where soil data are collected and prediction accuracy is evaluated. Zoning-based crop and fertilizer recommendations are validated against reference data and observed field conditions. The validation phase ensures system reliability through quantitative accuracy assessment and qualitative feedback on usability. Finally, the deployment and maintenance phase outlines potential large-scale application across Region˜10, including recommendations for system improvement, scalability, and long-term sustainability.

This study follows a modified waterfall research model to guide the development of the IoT-MCDA system shown in figure 4. The modified waterfall model allows an iteration to the previous phase for verification and validation, intended mainly for connecting stages. The process begins with the requirements gathering, in which the system will collect comprehensive data from the Department of Agriculture (DA) and credible online datasets. The datasets include zoning information and soil parameters, as well as crop-specific requirements for growth conditions. Moreover, all datasets will be integrated with the DA's zoning information.

It then proceeds to the requirements analysis, which identifies the need for soil health monitoring, crop prediction and fertilizer recommendation in the selected field in Region 10. A gradient boosting algorithm will be employed , using both sensor and zoning data inputs. Soil health prediction will classify soil quality levels, while crop yield forecasting will leverage an optimized machine learning model with metaheuristic-based feature selection to predict expected productivity. Fertilizer recommendations will be generated through a hybrid IoT-MCDA framework. Feedback in this stage may return to the requirements gathering phase if analysis reveals missing data, incorrect assumptions, or the need for additional crop or soil parameters to properly define system requirements.

The next phase is the system design, where the flow of the overall system is defined with detailed descriptions on how the system will work on implementing the IoT-based ML Integration system. This phase involves a comprehensive overview of the system design, architecture, the flowchart of the hardware and software that will be used in this system, as well as the web and mobile interfaces. The design ensures seamless integration of hardware components for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring control. The feedback may return to the requirements analysis stage if the proposed design cannot fully implement the defined requirements such as limitations in hardware integration or software module capabilities.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, in combination with multi-criteria decision analysis methods, and the configuration of hardware and software. Field nodes gather and preprocess soil data, then transmit it to a central server for storage, predictive analytics, and generation of crop suitability and fertilizer recommendations. A web dashboard visualizes soil health, farmer data, nutrient trends, weather, and zoning maps. The results in this stage can return to the system design phase if there are software or hardware integration challenges encountered, and it requires adjustments in design to ensure proper functionality.

Once completed, the testing phase is carried out in the selected area where soil data is gathered and prediction accuracy is measured. In this phase, the zoning-based recommendations are also validated. The validation phase then ensures reliability through quantitative accuracy testing and qualitative feedback on usability. Feedback from testing may return to the implementation phase to correct software bugs, or adjust hardware configurations for better performance before validation.

3.2 Requirements Gathering

The system requires comprehensive data collection to provide accurate, multi-cropping and fertilizer recommendation based on farming zoning in Budkinon, Philippines. Data will be sourced from the Department of Agriculture (DA), and credible online datasets. This includes detailed zoning information and soil parameters. To be included also are crop-specific requirements and growth conditions. Gathering these datasets ensures that the recommendation system is grounded in both environmental and agricultural best practices.

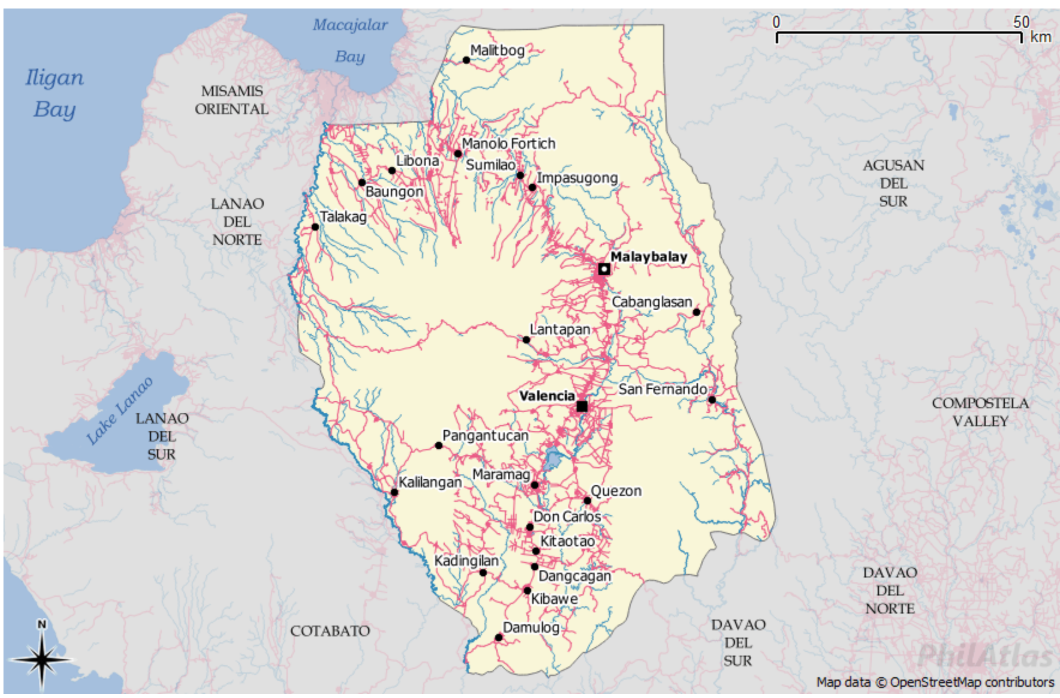
The types of data required encompass zoning information, and detailed crop profiles. Zoning data defines farming area, and land use classifications, while soil health parameters includes pH, NPK, salinity, moisture, and temperature. To be included as well is climate data such as historical rainfall, temperature, humidity, and forecast models which provide insights into environmental conditions that affect crop growth.

In preparation for system integration, the data will undergo a series of preprocessing steps. Missing sensor readings and data gaps will be addressed through interpolation, while noisy or incorrect data will be identified and removed to maintain integrity. Soil and climate parameters will be normalized into uniform ranges to enhance the performance of predictive models. Once cleaned and standardized, all datasets will be integrated with the Department of Agriculture's zoning information to ensure that recommendations are localized and relevant to the specific conditions in Bukidnon.

3.2.1 Research Setting

Figure 3.2

Research Setting (PhilAtlas)



The study will be conducted on specific farms in Region 10, specifically farms in Bukidnon, Philippines. The region is mostly agricultural, with vast lands focused on the cultivation of high-value crops such as Maize (Corn), Mungbean (Mongo), Peanut, Soybean, Squash (Kalabasa), Sweet Potato (Camote), Cassava, Taro (Gabi), Eggplant, Tomato, Pechay, and Cabbage. The farms in this region typically employ conventional and semi-modern farming practices, which offers a realistic setting to monitor and assess crop performance under actual farming conditions.

3.3 Requirements Analysis

The system requires the development and implementation of a machine learning component capable of performing soil health prediction, and crop recommendation. Machine Learning boosting algorithms will be employed for these tasks, using both sensor and zoning data. The sensors will collect important soil information such as Nitrogen (N), Phosphorus (P), Potassium (K), pH level, moisture, temperature, and salinity. At the same time, zoning data from the Department of Agriculture will provide information about the type of land, elevation, and climate conditions of the area. By combining these two data sources, the system can better understand the relationship between soil condition and the environment. Based on the collected data, the machine learning model will classify soil quality levels, while crop yield forecasting will leverage an optimized machine learning model by identifying patterns and relationships between soil parameters and environmental factors. To make predictions more accurate, the model will use a metaheuristic-based feature selection that focuses on the most relevant data and removes unnecessary or less useful information.

After analyzing the soil health parameters and producing crop recommendations, the system will generate fertilizer recommendations using a mix of IoT and Multi-Criteria Decision Analysis (MCDA). The MCDA method will help the system choose the best fertilizer type and correct amount for specific field conditions. To make sure the machine learning model produces reliable results, the dataset will be divided into three parts: 70% for training, 15% for validation, and 15% for testing. This ensures that the system learns effectively from the data, maintains accuracy, and avoids overfitting.

The system will undergo proper evaluation using both quantitative and qualitative approaches. Quantitative assessment will include metrics such as accuracy, precision, recall, and F1-score for soil health prediction. R^2, RMSE, and MAE for crop yield forecasting and consistency ratios and ranking stability indices for crop recommendation. Conducting these metric evaluations is important because they measure how well the machine learning model performs and how reliable its predictions are. By analyzing these values, the researchers can identify the model's strengths, detect errors, and make adjustments to improve performance before deployment. Cross-validation will ensure model robustness across different crop zones.Meanwhile, qualitative evaluation will involve focus group discussions and structured interviews with farmers and agricultural technicians in Region 10 to assess usability, reliability, and practical value. Feedback from these sessions will guide further refinement of the system to ensure its effectiveness in real-world agricultural applications.

3.3.1 User Definition

Following the requirements analysis, the system's users were defined as follows:

Farmer - Farmers are the primary beneficiaries of the system. Their farms provide localized data inputs through the soil sensor. In return, farmers receive soil health insights, and multi-cropping recommendations.

Department of Agriculture (DA) - The DA serves as both a data provider and a supervisory user of the system. They supply zoning information, and other agricultural zoning datasets that form the foundation of the recommendation system. Through the web-based dashboard, DA officials can visualize regional soil health trends, and monitor farming zones.

The specific user requirements are summarized in Table 1.

Table 2: User Requirements

User	Requirements
Farmer	- Access soil health insights (pH, moisture, nutrients) through a mobile app. - Receive crop suitability scores and multi-cropping recommendations. - Get fertilizer schedules tailored to soil condition and crop type.
Department of Agriculture	- Provide zoning maps, soil datasets, and crop profiles to the system. - Access a centralized web dashboard to visualize soil health, nutrient trends, and farmer data across Region 10.

This table summarizes the requirements of the system's two main users. Farmers need accessible insights and recommendations to guide timely crop and fertilizer decisions, while the Department of Agriculture requires integrated data and analytics to support monitoring, reporting, and regional agricultural planning.

3.3.2 System Requirements

The system requirements shown in Table 2, were categorized into Input, Process, Output, Control, and Performance specifications to aid the system design and implementation.

Table 3: System Requirements

Category	System Requirement
Input Requirements	- The system shall accept input data directly from farmers and agricultural authorities through the designated interface. - The system shall acquire and process soil data from integrated sensors or external data sources. - The system shall accept soil health data from the sensor as input for hardware operation. - The system shall retrieve weather information from external APIs. - The system shall define sensor communication parameters to ensure consistent and accurate data acquisition from IoT nodes.
Process Requirements	- The system shall combine weather, soil, and farmer-provided data into a single decision-making framework. - The system shall employ boosting algorithms to enhance the accuracy of predictive crop recommendations. - The system shall analyze data-driven insights to balance yield potential with resource availability. - The system shall determine an appropriate planting timeline based on environmental conditions and crop growth requirements.
Output Requirements	- The system shall present crop recommendations via a visual interface on mobile and web applications. - The system shall display multi-cropping options, which shows compatible crop pairings. - The system shall present localized weather forecasts. - The system shall provide soil health reports with clear visual indicators. - The system shall display information on current crops cultivated by the farmer.

Category	System Requirement
	- The system shall display farm details of the farmer. - The system shall display the locations and overview of registered farms. - The system shall display a list of all farmers registered. - The system shall present crop rules, including crop rotation guidelines, and companion planting practices.
Control Requirements	- The system shall provide secure login and authentication for farmers and agricultural authorities. - The system shall restrict access to sensitive data based on user roles and permissions. - The system shall allow administrators to update crop rules and farm locations. - The system shall protect sensor and farmer data during transmission, and storage using standardized encryption methods and user authentication protocols.
Performance Requirements	- The system shall scale efficiently to accommodate growth in registered farmers and farms without requiring major redesign. - The system must be able to have offline syncing for farmer features. - The system must provide accurate crop recommendations for farmers. - The system shall synchronize data reliably between mobile and web platforms.

The system requirements were defined to ensure the effective collection, processing, and presentation of farm and soil data to support informed crop management decisions. The primary objective of the system is to assist farmers by integrating farm-specific information and soil health metrics to generate accurate and actionable crop and fertilizer recommendations. Input requirements focus on collecting data from farmers and IoT-based sensors, while process requirements involve the use of advanced machine learning techniques, particularly gradient boosting algorithms, to analyze soil conditions, predict crop suitability, and estimate potential yield performance. Outputs are delivered through mobile and web applications to provide farmers and administrators with clear, visualized recommendations and reports. Control requirements ensure secure access through role-based permissions and administrative functions for managing crop data and farm records. Lastly, performance requirements address scalability, offline data synchronization, and recommendation accuracy to ensure reliable decision support across diverse agricultural environments.

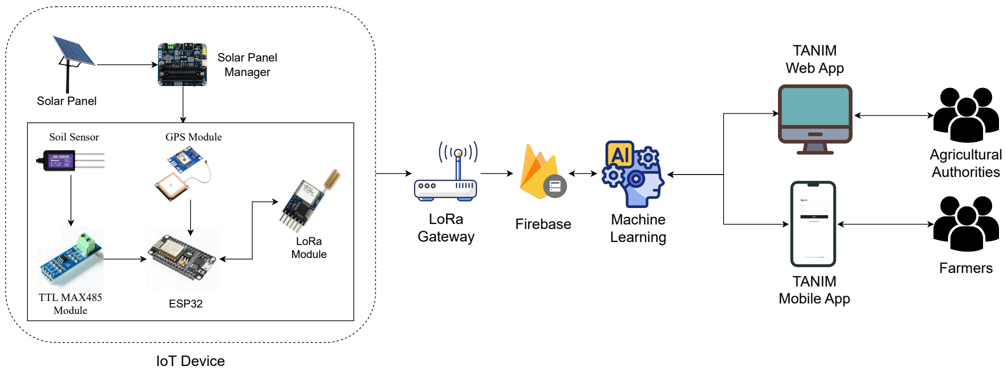
3.4 System Design

In this section, the flow of the overall system is defined with detailed descriptions on how the system will work on implementing the IoT-based ML integration system. This involves a comprehensive overview of the system design and architecture, and the flowchart of the hardware and software that will be used in this system. The design ensures seamless integration of hardware components for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring and control.

3.4.1 System Architecture

Figure 3.3

System Architecture



The figure \ref{fig:SystemArchitecture} illustrates the system architecture of the proposed Soil Health Monitoring, Fertilizer, and Crop Recommendation System. The system will begin with a solar-powered IoT sensing unit deployed in the field. A solar panel, regulated by a solar charge controller, will supply continuous power to the device, enabling reliable operation even in remote agricultural areas. In addition to powering the sensing unit, the energy subsystem will continuously monitor the battery level to ensure uninterrupted operation and early detection of power-related issues.

At the core of the sensing unit will be a soil sensor designed to measure key parameters such as NPK, pH, salinity, moisture, and temperature. Since the sensor outputs RS-485 differential signals, a MAX485 module will be used to convert these signals into TTL levels readable by the microcontroller. The MAX485 is selected for its ability to support long-distance and noise-resistant communication, which is essential in outdoor environments where sensor cables may be long and exposed to interference. Its low power consumption further supports the solar-powered architecture of the system.

The ESP32 microcontroller will serve as the main processing unit. It is chosen for its high processing capability, dual-core architecture, low power consumption, and support for multiple communication protocols. In addition to collecting soil sensor readings, the ESP32 will also monitor the battery voltage of each sensing probe through an analog input connected to a voltage divider circuit. This allows the system to estimate the battery charge level in real time. The ESP32 will perform filtering and averaging on both sensor and battery readings to minimize noise and transient fluctuations.

Although the sensor provides direct measurements, these readings may fluctuate due to electrical noise, environmental interference, soil inconsistencies, or transient signal spikes. Filtering will remove noise and outliers, while averaging multiple samples will reduce jitter and produce stable, reliable values. The same data refinement approach will be applied to battery measurements to ensure accurate reporting of device power status. This processed data, including soil parameters and corresponding battery status per probe—will then be packaged for transmission.

For long-range data transmission, the system will integrate a LoRa module. LoRa technology will enable low-power, long-distance communication, making it suitable for large agricultural fields where Wi-Fi or cellular connectivity may be limited. Sensor data packets transmitted via LoRa will include soil measurements, device identifiers, timestamps, and battery status information for each probe, allowing remote monitoring of both soil health and device power conditions.

The LoRa gateway receives the transmitted packets and forwards the data to the Firebase cloud database via Wi-Fi connectivity. Firebase will serve as the cloud database, providing real-time storage and retrieval of sensor data and battery status logs. Once received in the cloud, the data will undergo preprocessing and validation before being passed to the machine learning model. The machine learning engine will analyze soil patterns, detect nutrient or moisture deficiencies, and generate fertilizer recommendations, crop suggestions, and soil health indicators based on learned relationships from historical and real-time data, while battery data will be used for system health monitoring and maintenance planning.

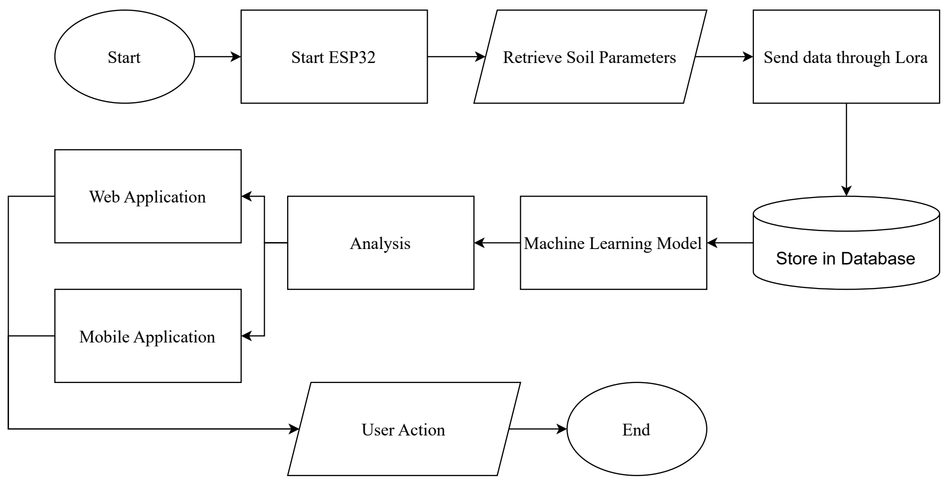
System outputs will be delivered through the TANIM Web App and the TANIM Mobile App. The Web App will provide agricultural authorities with dashboards, maps, and analytical tools for monitoring soil conditions and device status across regions. The Mobile App will be designed for farmers, presenting information in a clear and accessible format, including soil health insights, fertilizer recommendations, crop suggestions, battery status indicators, and critical alerts such as low-battery warnings. Farmers will also be able to submit feedback through the app regarding crop performance and field practices. This feedback, combined with continuous sensor and battery data, will be incorporated into the machine learning workflow to enhance the accuracy and reliability of future recommendations.

Through the integration of IoT hardware, renewable energy, cloud services, and machine learning, the proposed system will support data-driven decision-making for both farmers and agricultural authorities, while also enabling proactive monitoring of device power health, ultimately contributing to improved crop yield and sustainable farming practices.

3.4.2 Flowchart

Figure 3.4

System Flowchart



The figure 3.4 presents the operational workflow of the proposed soil health monitoring and recommendation system, highlighting the data flow from the field device to the end-user applications. The process initiates at the sensing stage, where the ESP32 microcontroller, powered through a solar energy management unit, interfaces with soil sensors to capture essential soil parameters, including Nitrogen, Phosphorus, and Potassium (NPK), pH, moisture, temperature, and soil salinity. These parameters are critical indicators of soil fertility and crop suitability. Once collected, the data is processed locally by the ESP32 to ensure efficient packaging before transmission. Leveraging LoRa technology, the device transmits the data wirelessly over long distances to a gateway, which serves as the intermediary link between the sensing unit and the cloud infrastructure. The gateway then forwards the collected data to a cloud database, ensuring reliable and scalable data storage for subsequent analysis.

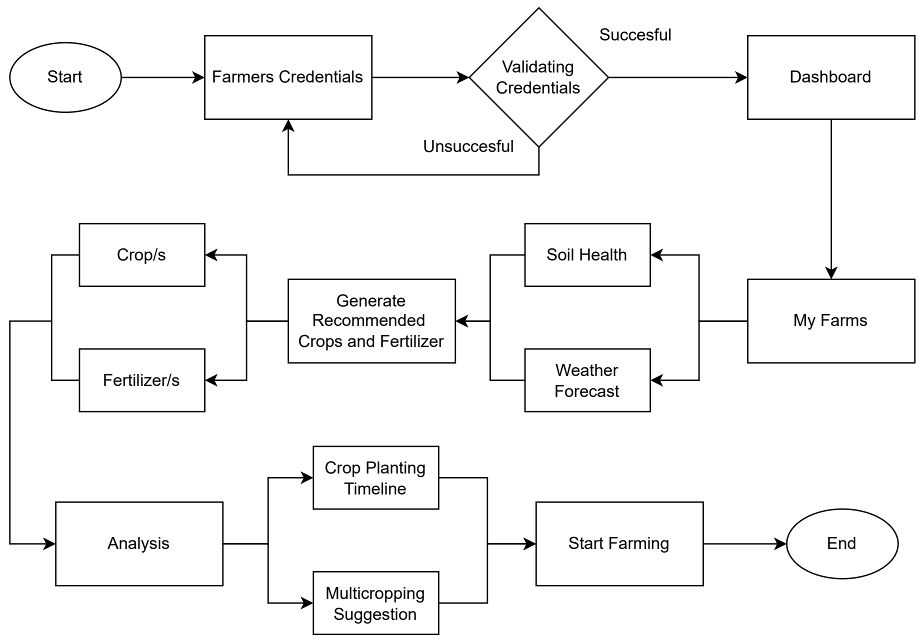
Within the cloud environment, the raw soil data undergoes processing and analysis through machine learning models. These models are trained on historical soil and crop datasets to identify patterns, classify soil conditions, and predict nutrient deficiencies. By transforming raw sensor readings into interpretable outputs, the system generates actionable insights regarding soil health and crop management. The analytical results include fertilizer recommendations and crop suitability assessments, enabling evidence-based decision-making for end users. This layer of intelligence ensures that data collected from the field is not merely stored but is also transformed into knowledge that directly supports precision agriculture practices.

The processed insights are subsequently disseminated through two primary platforms: a web application and a mobile application. The web application is designed for agricultural authorities and extension officers, providing comprehensive monitoring dashboards, spatial analysis tools, and regional soil health reports. This enables decision-makers to track soil conditions at scale, identify priority intervention areas, and develop region-specific agricultural strategies. The mobile application targets farmers as the primary stakeholders, offering a user-friendly interface to access real-time soil health status, recommended fertilizer dosages, and crop-specific advisories. Notifications and alerts are integrated into the system to provide timely guidance, particularly in scenarios requiring immediate action.

An integral component of the workflow is the feedback mechanism incorporated within both platforms. Farmers and users are able to submit observations, report crop outcomes, and validate system recommendations. This feedback not only enhances farmer engagement but also provides valuable data for iterative improvements of the machine learning models. As a result, the system establishes a continuous learning cycle wherein recommendations are refined over time, leading to greater accuracy and contextual relevance. Overall, the workflow described in Figure 3.4 demonstrates how the integration of IoT sensing, cloud computing, and machine learning facilitates a comprehensive, data-driven approach to soil health management and crop productivity enhancement.

Figure 3.5

Software Mobile Application



The figure 3.5 illustrates the workflow of the farming recommendation system, highlighting the interaction between farmers and the application in obtaining crop suggestions tailored to soil and weather conditions. The process begins with user authentication, wherein the farmer logs into the system using their credentials. The application validates these credentials against the stored records in the database, ensuring secure access. If the credentials are incorrect, the system prompts the user to re-enter valid information, thereby safeguarding the platform against unauthorized access. Upon successful login, the farmer is redirected to the dashboard, which serves as the central interface for accessing personalized farm data and system functionalities.

From the dashboard, the farmer navigates to the “My Farms” section, where the application consolidates and displays farm-specific data. This includes soil health parameters gathered from IoT-enabled sensors and environmental information derived from integrated weather forecasting services. The system then processes this combined dataset using its analytical and decision-making algorithms to generate a list of recommended crops that are most suitable for the prevailing soil and weather conditions. By leveraging both real-time and contextual data, the system ensures that the recommendations are accurate, adaptive, and relevant to the farmer’s specific location and situation.

In areas where internet connectivity is limited, the application supports an offline mode that allows farmers to continue accessing previously stored data and recommendations. This feature ensures uninterrupted usability by caching essential farm information, soil records, and the most recent crop recommendations on the device. Once the connection is restored, the system automatically synchronizes any new data and updates the stored records in the cloud. Through this mechanism, farmers in remote areas can still make informed decisions and manage their farms effectively, even without continuous internet access.

Once the recommendations are presented, the farmer can review the options and select the crop that aligns with their goals, preferences, and available resources. Following this selection, the system provides advanced features designed to further support farm management. These include multicrop suggestions, which propose crop diversification strategies for risk mitigation and sustainable land use, as well as a detailed crop planting timeline that outlines optimal planting and cultivation schedules. Such features are aimed at enhancing farming efficiency, improving resource allocation, and maximizing yield potential.

The final stage of the workflow involves the farmer applying the recommendations provided by the system. By combining data-driven crop selection with structured guidance on crop management, the system enables farmers to initiate the farming process with greater confidence and scientific support. This workflow underscores the system's capacity to guide users from initial login and farm data access to informed decision-making and implementation in the field. Ultimately, the integration of secure user access, real-time soil and weather data, intelligent recommendation algorithms, and decision support tools demonstrates how the farming recommendation system contributes to improved agricultural productivity and long-term sustainability.

Figure 3.6

Web Application Flowchart

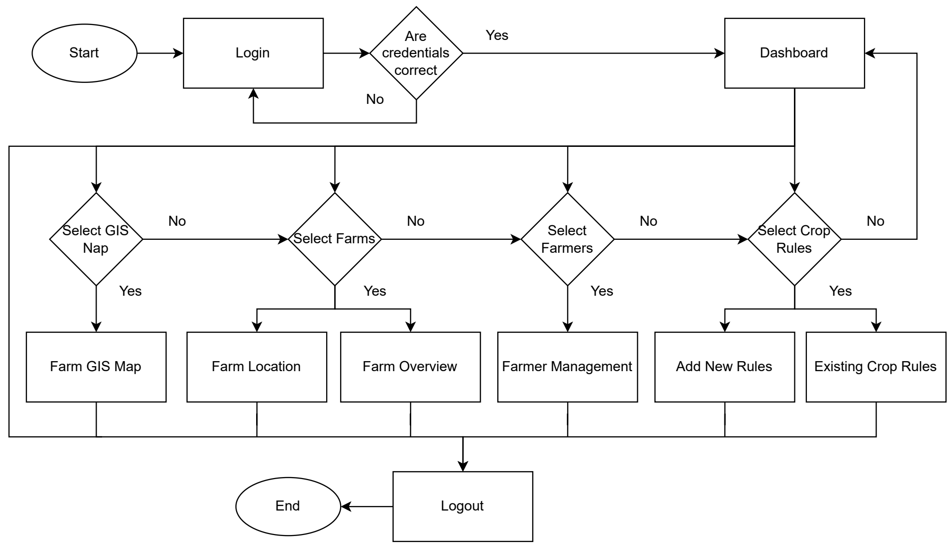


Figure 3.6 illustrates the workflow of the Farm Management Web Application, which is designed to support agricultural administrators in managing and monitoring farming activities at a regional scale. The process begins with a secure login mechanism, where the administrator is required to enter valid credentials. The system verifies these credentials against stored records to ensure authorized access. If the credentials are invalid, the administrator is redirected back to the login page to re-enter the correct information. Upon successful authentication, the administrator is directed to the dashboard, which serves as the central navigation interface of the system.

From the dashboard, the administrator can access four primary modules represented by the decision nodes in the flowchart. The GIS Map module provides geospatial visualization of registered farms, enabling spatial analysis and regional monitoring. The Farms module supports the management of farm-related information, including farm locations and farm overviews, ensuring accurate documentation of farm records. The Farmers module focuses on farmer management, allowing administrators to organize farmer profiles and maintain farmer-related data. The Crop Rules module facilitates the administration of agricultural policies by enabling the addition of new crop rules and the review of existing crop regulations.

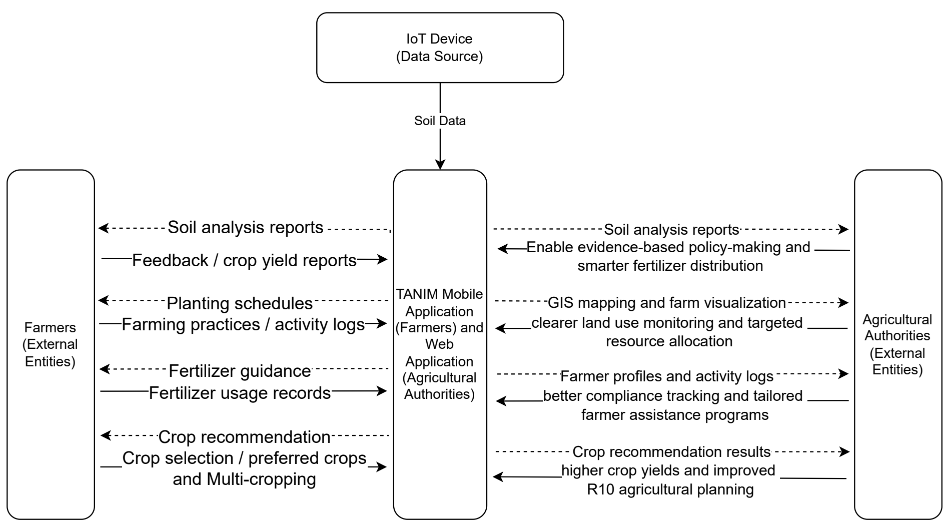
Each module is seamlessly integrated with the dashboard, allowing administrators to navigate efficiently between system functions. After completing tasks within a selected module, the administrator may return to the dashboard or securely terminate the session through the logout function. This workflow highlights the system's structured approach to access control, centralized administration, and policy management, supporting effective monitoring, data-driven decision-making, and sustainable agricultural governance for the Department of Agriculture.

3.4.3 Context Level Diagram

The Context-Level Diagram provides a high-level view of how the TANIM system interacts with its primary stakeholders: Agricultural Authorities and Farmers. It illustrates how data flows into and out of the system without detailing internal processes.

Figure 3.7

Context-Level Diagram Between the System And The External Entities



The Context-Level Diagram illustrates the integrated interaction of the TANIM system with both Agricultural Authorities and Farmers through its unified web and mobile applications. Agricultural Authorities do not directly generate raw data, instead, they receive processed insights derived from IoT devices deployed in the field, including soil analysis reports, GIS-based farm visualizations, farmer profiles, and crop recommendation results. These outputs support evidence-based policymaking, targeted resource allocation, compliance monitoring, and improved agricultural planning. At the same time, IoT devices serve as primary data sources by providing soil and environmental information to the system, while farmers contribute essential inputs such as crop yield feedback, farming activity logs, fertilizer usage records, and crop preferences. TANIM processes this continuous exchange of data to produce actionable outputs such as planting schedules, fertilizer guidance, crop recommendations, and farming practice records. Through this unified data flow, the system transforms raw field data into actionable intelligence that supports sustainable farming practices, enhances farmer productivity, and enables informed, timely decision-making that contributes to national food security goals.

3.4.4 Data Flow Diagram

The Data Flow Diagram visually represents how data flows within the TANIM system. It highlights the interactions between IoT devices, farmers/agricultural authorities, and the system's core processes, demonstrating how data is retrieved, stored, and analyzed to provide soil health insights and crop/fertilizer recommendations.

Figure 3.8

Data Flow Diagram

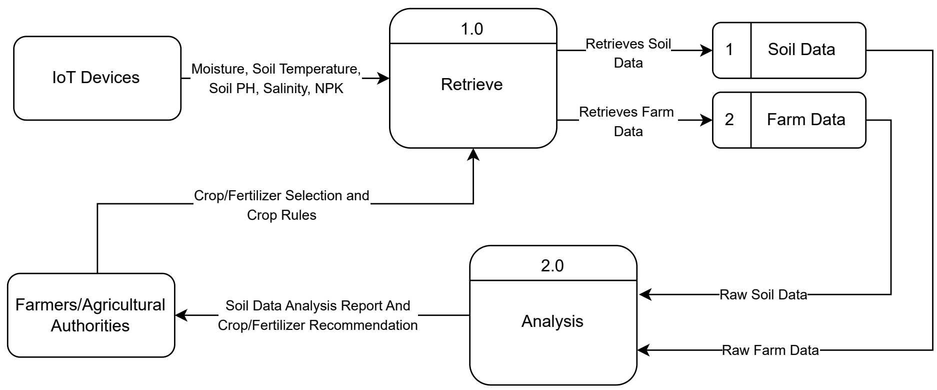


Figure 3.9 shows the Data Flow Diagram (DFD) of the TANIM system. The data flow between the user and the system is divided into two main processes which is the retrieval and analysis process. The “retrieve” process is responsible for collecting soil and farm data from IoT devices, including sensor readings of NPK levels, pH, moisture, salinity, and temperature, as well as crop and fertilizer inputs provided by the users. These data are validated, preprocessed, and stored in the system’s database to ensure consistency, accuracy, and ease of access for subsequent processing.

In the “analysis” process, the system applies machine learning algorithms to evaluate the stored data, identifying patterns and relationships among soil properties, environmental factors, and crop performance. The outputs of this analysis include predictions on soil health status, crop suitability, and potential yield. Based on these predictions, the system generates tailored crop and fertilizer recommendations that consider both environmental conditions and expected productivity, enabling farmers to make informed decisions that optimize resource utilization.

Finally, the processed results are delivered to farmers and agricultural authorities through a user-friendly interface in the form of soil data analysis reports, visual dashboards, and actionable recommendations. These outputs facilitate practical decision-making, support strategic crop planning, and contribute to more efficient and sustainable agricultural practices across Region 10.

3.4.5 Use Case Diagram

The Use Case Diagram illustrates how different actors interact with the TANIM system. It presents the system's primary functionalities and shows how users access and utilize agricultural data for farm management and monitoring. The diagram emphasizes the roles of both Farmers and the Department of Agriculture in supporting efficient and data-driven agricultural practices.

Figure 3.9

Use Case Diagram

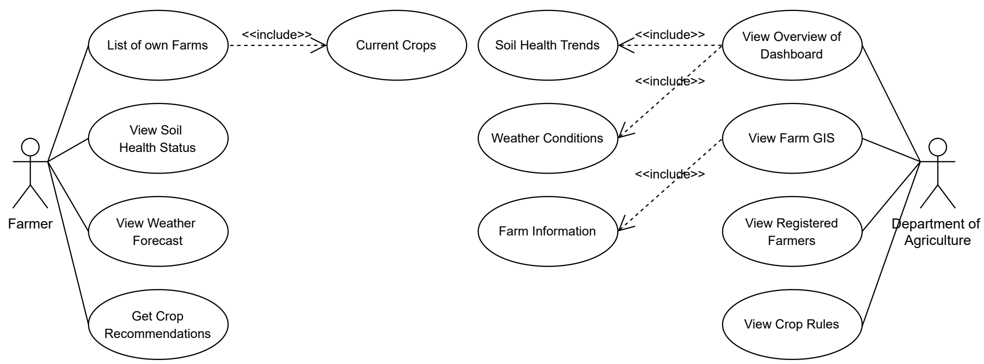


Figure 3.9 shows the interactions of the Farmer with the system. Farmers can view a list of their own farms, which includes information about the current crops being cultivated. This feature helps farmers organize and monitor their farming activities. Farmers are also able to view the soil health status of their farms, allowing them to assess land condition and make informed decisions regarding crop management. In addition, the system provides access to weather forecasts, enabling farmers to plan daily operations and prepare for potential weather-related risks. The system also offers crop recommendations tailored to soil health data, weather conditions, and existing farm information to support improved agricultural productivity.

The figure also illustrates the interactions of the Department of Agriculture with the system. The Department can access an overview dashboard that presents soil health trends and weather conditions, providing insights into agricultural conditions across different farms. Farm information is integrated into the dashboard to support monitoring and analysis. The Department can also view farm locations through the Farm GIS feature, enabling spatial visualization of agricultural areas. Furthermore, the system allows the Department to view registered farmers and oversee crop rules and regulations, ensuring compliance with established agricultural policies and standards.

The Use Case Diagram demonstrates how TANIM supports both farm-level management and institutional oversight. By providing relevant tools and information to different users, the system enhances coordination, monitoring, and decision-making in agricultural management.

3.4.6 Use Case Table

The Use Case Table provides a structured and detailed representation of the functionalities presented in the Use Case Diagram. It enumerates each system function, identifies the primary actor involved, and describes the purpose of each interaction. In addition, the table highlights relationships among use cases through included or related functionalities.

The use cases associated with the *Farmer* focus on farm-level management and decision support. These include viewing a list of registered farms and current crops, monitoring soil health status, accessing weather forecasts, and receiving crop recommendations. These functionalities enable farmers to effectively manage their farming activities, assess environmental conditions, and make informed decisions to improve agricultural productivity. Certain use cases depend on related system features, such as soil health and weather data, which support the generation of crop recommendations.

The use cases associated with the *Department of Agriculture* emphasize monitoring, analysis, and regulatory oversight. These include viewing an overview dashboard that integrates soil health trends, weather conditions, and farm information, accessing farm locations through the Farm GIS feature, and viewing registered farmers. The Department may also view crop rules to ensure compliance with agricultural standards and policies.

Table 4: Primary Use Cases of the TANIM System

Use Case Name	**Description**	**Primary Actor**	
		Farmer	**Department of Agriculture**
List Own Farms	Enables farmers to view their registered farms.	✓	
View Soil Health Status	Allows users to view soil condition data such as nutrients, fertility, and health indicators.	✓	
View Weather Forecast	Provides weather forecasts relevant to farm locations.	✓	
Get Crop Recommendations	Provides crop recommendations based on soil health, weather, and farm data.	✓	
View Dashboard Overview	Allows users to view a summarized overview of agricultural data for monitoring and analysis.		✓
View Farm GIS	Displays geographic locations of farms through a GIS interface.		✓
View Registered Farmers	Allows the Department of Agriculture to view all registered farmers in the system.		✓
View Crop Rules	Allows the Department to view and manage crop rules and regulations.		✓

Table 5: Included Use Cases of the TANIM System

Use Case Name	**Description**	**Primary Actor**	
		Farmer	**Department of Agriculture**
View Current Crops	Allows farmers to view crops currently cultivated on their farms.	✓	
View Soil Health Trends	Enables monitoring of soil health changes over time across regions.		✓
View Weather Conditions	Allows the Department to view current and historical weather data.		✓
View Farm Information	Enables the Department to access detailed information about farms.		✓

3.4.7 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the structural design of the database system that supports the TANIM platform. It defines the relationships between the core entities—Department of Agriculture, Farmer, Farm, Crop Recommendation, and Crops—and outlines how data is organized to support agricultural management and decision-making processes. The diagram emphasizes data integrity, relational links, and the flow of information between administrative, user, and farm-level components.

Figure 3.10

Entity Relationship Diagram

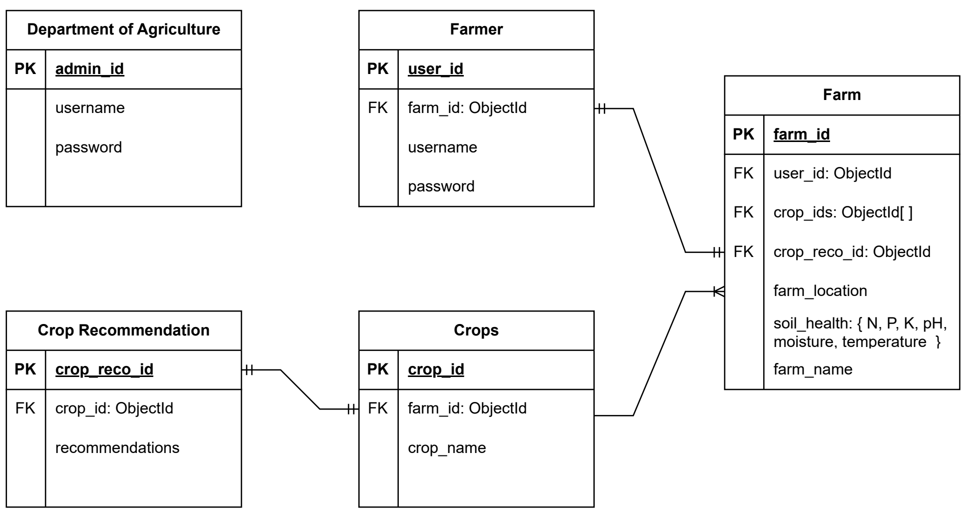


Figure 3.10 shows the core entities and their relationships within the system. The *Department of Agriculture* entity acts as the administrative unit, storing credentials for system oversight and policy management. The *Farmer* entity represents system users, each linked to one or more farms through

the foreign key `farm_id`. The *Farm* entity serves as the central data hub, containing identifiers, location data, and a structured `soil_health` attribute that records key agronomic parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, moisture, and temperature. Each farm is associated with multiple crops through an array of `crop_ids` and is linked to a customized *Crop Recommendation* based on real-time environmental and soil data.

The *Crop Recommendation* entity stores data-driven advisories (`recommendations`) that are linked to specific crops via `crop_id`. The *Crops* entity catalogs available crop types and references the farms where they are cultivated. These relationships ensure that farm data, crop information, and recommendations are logically connected, enabling efficient queries and consistent data retrieval.

The ERD demonstrates how TANIM structures agricultural data to support both operational and strategic functions. By maintaining clear relational links between farmers, farms, crops, and recommendations, the system ensures data coherence, supports scalable analysis, and facilitates informed agricultural decision-making at both individual and institutional levels.

3.4.8 Hardware Design Flowchart

Figure 3.11

Hardware Design Flowchart

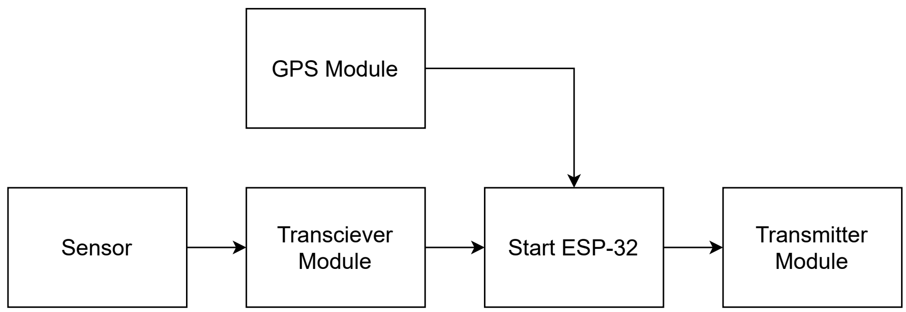


Figure \ref{fig:HardwareFlowchart} illustrates the workflow of the hardware components. The process begins with the initialization of the ESP32 microcontroller, which serves as the central control unit of the system. The GPS module provides location data to the ESP32, while the sensor module collects environmental data required for monitoring.

Once the sensor data is acquired, it is passed to the transceiver module, which facilitates communication between the sensing unit and the ESP32. The ESP32 processes the received data and forwards it to the transmitter module. Finally, the transmitter module sends the processed data to the designated external system, completing the hardware operation cycle.

The flowchart represents a sequential hardware process in which each stage depends on the successful execution of the previous step. The ESP32 coordinates sensor data acquisition, GPS integration, and communication control, while the transceiver and transmitter modules ensure reliable data transmission.

3.4.9 3D Design

Figure 3.12

3D Container Design

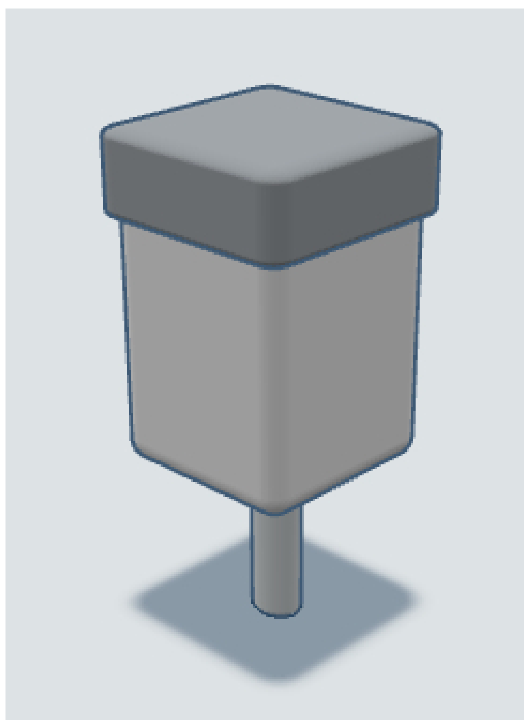


Figure 3.12 displays the 3D container design that will be 3D printed for the container of the hardware. This will contain the hardware components which are etched in a PCB board that will be used in this system. The con-

tainer should be durable to ensure that it can secure the hardware components inside of it.

3.4.10 Materials and Cost

Table 6: Hardware Components and Cost

Component	Price
ESP32 Microprocessor	₱150
Soil Sensor	₱1,654.90
Solar Power Manager	₱860
Solar Panel	₱320
Lithium Ion Battery	₱120
GPS Module	₱255
Transceiver Module	₱163
Transmitter Module	₱180
Miscellaneous (Cable, Mounts)	₱1,500
Total	**₱5,202.90**

3.5 Implementation

The Implementation phase involves developing the software and hardware components. Each function was ensured to function according to the design specifications.This phase intends to transform the conceptual design into a working prototype that can perform data acquisition, processing, and output for TANIM.

3.5.1 Software Implementation

The software implementation for TANIM is designed to provide an efficient, reliable, and user-friendly platform that caters to the needs of both the Farmer and the Department of Agriculture. To ensure optimal accessibility and performance, the frontend for farmers is developed using React Native, a framework well-suited for creating responsive and cross-platform mobile applications. The mobile application also features offline data handling, allowing farmers to access previously synced soil and crop data even without an internet connection. Any new inputs or updates made while offline are temporarily stored locally on the device and automatically synchronized with the cloud once connectivity is restored. This ensures uninterrupted functionality in rural areas with limited network coverage.

Meanwhile, the frontend for the Department of Agriculture is implemented using React.js, providing a seamless and interactive web application experience. The web application integrates advanced analytics and visualization tools, enabling agricultural authorities to interpret soil data, monitor farm conditions, and analyze regional agricultural trends through interactive charts, dashboards, and geospatial maps. These insights support data-driven planning, resource allocation, and policy formulation.

Both user interfaces interact with a shared backend developed in Flask, which centralizes data management and ensures consistent functionality across platforms. This architecture enables efficient communication between the mobile and web applications while supporting scalability, maintainability, and ease of integration with additional features or services in the future.

3.5.2 Hardware Implementation

The hardware implementation of the TANIM system is centered on an IoT-based device designed to collect soil data and send it to the cloud for processing and analysis. The core of the device is the ESP32 microcontroller, selected for its high processing capability, built-in Wi-Fi and Bluetooth connectivity, low power consumption, and compatibility with multiple sensor modules. These features make it ideal for continuous field monitoring and data transmission in remote agricultural areas.

A soil sensor is used to measure key soil parameters such as pH, NPK, moisture, temperature, and salinity. This sensor was chosen because it provides accurate and real-time soil data essential for evaluating soil health and guiding fertilizer and crop recommendations.

3.5.3 Machine Learning Model

The TANIM system will use a boosting algorithm as the core of its predictive model, chosen for its efficiency, scalability, and high accuracy in handling large and complex datasets. This machine learning approach analyzes multiple factors, including soil type, environmental conditions, and historical yield data, to generate accurate predictions of soil health, crop suitability, and potential yield. By leveraging the predictive power of boosting algorithms, the system is able to deliver personalized and data-driven crop and fertilizer recommendations that optimize productivity and promote efficient resource utilization.

3.5.4 Security

To enhance the security of the TANIM system, Employ Role-Based Access Control (RBAC) will be strictly enforced at the API level so a Farmer cannot accidentally or maliciously access DA-level administrative rules. Two-Factor Authentication (2FA) will also be implemented during the login process for both Farmers and Agricultural Authorities. After entering valid credentials in the mobile or web application, users will be required to provide a secondary verification code sent to their registered email or mobile number. This adds a critical layer of protection against unauthorized access to sensitive farm data and regional soil health insights.

The TANIM system secures the exchange of information across multiple layers of the architecture. For the long-range link between the field sensors and the gateway, AES-128 encryption is a standard feature of LoRa technology that is utilized to ensure that soil parameters remain confidential during wireless transmission. Furthermore, communication between the Flask backend and the user applications is conducted over HTTPS (TLS/SSL) to prevent data interception or man-in-the-middle attacks. This ensures that real-time insights, such as fertilizer recommendations and crop suitability scores, are delivered securely to the end-user.

Data integrity begins at the collection point, where the ESP32 microcontroller handles localized processing and filtering to prevent signal spikes from being recorded as valid data. The hardware is protected by a durable 3D-printed container, designed to secure the PCB and sensors from physical tampering in remote agricultural settings. On the cloud side, Firebase Security Rules are configured to ensure data isolation; Farmers are only permitted to write data associated with their specific accounts, while the Department of Agriculture is granted read-only access to regional soil health trends for monitoring purposes. Regular database backups and encryption-at-rest further safeguard the repository of environmental and farmer-defined parameters.

For wireless communication, the system uses a LoRa (Long Range) module, selected for its ability to transmit data over several kilometers with very low power consumption. This makes it suitable for rural agricultural zones with limited internet access or unstable connectivity.

The entire system is powered by a solar panel managed through a solar panel controller, ensuring a sustainable and uninterrupted power source even in off-grid farming locations. The modular design of the hardware also makes the system scalable and adaptable, allowing additional sensors or communication modules to be integrated easily as agricultural requirements evolve.

3.5.5 Data Acquisition and Storage

The sensor is tasked with acquiring data by measuring a variety of critical indicators of soil health. The collected data is then sent to a cloud storage, which provides a secure, scalable, and easily accessible repository for large volumes of environmental data. This cloud-based storage solution not only ensures data integrity and reliability but also facilitates seamless integration with processing stages. The stored data is then utilized as input for the trained machine learning model, which analyzes the information to generate precise insights, predictions, and recommendations regarding soil conditions. Overall, this supports informed decision-making for agricultural management.

3.5.6 Machine Learning Model

The TANIM system will use a boosting algorithm as the core of its predictive model, chosen for its efficiency, scalability, and high accuracy in handling large and complex datasets. This machine learning approach is integrated with Multi-Criteria Decision Analysis (MCDA) methods to enhance decision-making by considering multiple factors such as soil type, weather conditions, and historical yield data. The machine learning model first analyzes and predicts soil health and potential crop performance based on numerical and environmental data. The results from the machine learning model, such as soil quality scores and yield probabilities, are then used as inputs for the MCDA process. The MCDA evaluates these outputs together with additional factors like crop suitability, fertilizer requirements, and environmental sustainability to rank and recommend the best crops for a specific location. By combining the predictive capability of machine learning with the structured evaluation of MCDA, the system produces accurate, context-aware, and data-driven crop recommendations that help optimize productivity and support sustainable farming practices.

3.6 Test and Evaluation

Testing and evaluation will be conducted to verify the system's effectiveness, accuracy, and usability for crop recommendation and soil health monitoring. This process will be designed to ensure that the hardware components, software components, and machine learning models will function according to what we aim for this study, which involves hardware testing and calibration, functional testing, system integration testing, performance testing, and usability testing. Hardware testing and calibration will focus on verifying the accuracy of the sensors to ensure reliable soil data. Functional testing will examine whether the mobile and web applications operate correctly and deliver the expected outputs of each feature. To ensure that the hardware, cloud database, machine learning, and applications works together with proper data flow and error handling, system integration will be implemented. For measuring the responsiveness of the system and to evaluate the system's predictive accuracy of the machine learning models using various performance metrics we will be conducting performance testing. Moreover, usability testing will assess the practicality and user-friendliness of the system by involving farmers and agricultural experts in field trial and feedback collection.

3.6.1 Hardware Testing and Calibration

Hardware testing and calibration will ensure that the IoT sensors will provide accurate measurements of soil health parameters such as nitrogen, phosphorus, potassium, pH, moisture, salinity, and temperature. To conduct this testing, the sensors will be deployed in field testing on the actual farms in Region 10, also to ensure consistency under varying field conditions the readings will be recorded at different time and day. Cross checking will be applied by using the Department of Agriculture's soil testing laboratory and soil test kits of the same soil samples from the same farms to ensure reliability and accuracy.

Through statistical measures such as Mean Absolute Error (MAE) the sensor readings will be compared against laboratory results. a maximum allowable error threshold of ±5 will be set, if the difference between the sensor and laboratory values exceeds this threshold error, calibration adjustments will be applied to the sensors until acceptable accuracy will be achieved. It will be calculated by:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{1}$$

This process ensures that the data generated by the hardware is reliable enough to be used in machine learning models and the recommendation system.

3.6.2 Functional Testing

Functional testing involves verifying whether each feature of the mobile and web applications operates correctly according to the system design. For the mobile application, the following features will be tested: secure login and authentication, retrieval of soil health readings, display of crop recommendations for multi-cropping, fertilizer optimization, and offline data synchronization. For the web application, features such as farm visualization, farmer profile management, and crop rule configuration will be tested.

In addition, synchronization between the database, mobile, and web applications will be tested to ensure that all data are accurately updated and consistent across platforms. This includes verifying that soil readings, user information, and crop recommendations are properly synchronized when data are sent or retrieved from the cloud.

Both black-box testing and white-box testing approaches will be applied to evaluate the system's functionality. Black-box testing will focus on verifying the system's external behavior without examining its internal code. Testers will input various data to check if the outputs match the expected results, ensuring that all functions operate as intended from a user's perspective. Meanwhile, white-box testing will be conducted by examining the internal logic, data flow, and code structure of the applications. This will help identify hidden logical errors, broken functions, or inefficient code implementations.

Stress testing will also be performed by simulating multiple user logins and concurrent farm entries to evaluate system performance under heavy load. Through these testing activities, the TANIM system aims to ensure accurate functionality, reliable synchronization between components, and stable performance in real-world agricultural applications.

3.6.3 System Integration Testing

System integration testing will evaluate whether the hardware, software, cloud database, and machine learning model communicates seamlessly with each other as a single system. The system integration testing will be tested by following the full flow of the system: soil health readings captured by the sensor and processed by the ESP32 microcontroller, then transmitted by via the LoRa module to the gateway, forwarded to the cloud database, then processed by the machine learning models, and finally displayed on the mobile and web applications.

This process confirms that the system works together as one reliable system. Furthermore, tests will be conducted to observe how the system handles incomplete or corrupted data to ensure that it can manage error to prevent misleading recommendations to farmers.

3.6.4 Performance Testing

For the system level performance, it is to determine if the system could handle multiple data acquisition and provide timely feedback to the users. The data transmission success rate will be monitored by comparing the successful sensor to cloud uploads against the attempted uploads over multiple trials. Latency tests measure the time from the sensor data capture to its display in the mobile and web application, with an acceptable range set to 30 seconds. Scalability will be evaluated by simulating multiple farms connected simultaneously to ensure that the cloud database and applications could handle increased workloads without delays and crashes. These will ensure that the system is reliable and efficient under real-world agricultural conditions where multiple users operate.

For the machine learning evaluation, it verifies the predictive capability of the machine learning models that will be used on crop recommendation for multi-cropping and fertilizer optimization. The dataset will be split into 70-15-15 for training, evaluation, and testing for maximizing the predictive accuracy and avoiding overfitting.

For the crop recommendation, the machine learning models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices to determine the effectiveness of the system. The following formulas are going to be use:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \tag{6}$$

These evaluation steps confirm that the machine learning models provide scientifically sound predictions and deliver crop and fertilizer recommendations that are consistent, reliable, and effective under different scenarios.

3.6.5 Usability Testing

The usability testing will assess the practicality, accessibility, and user satisfaction of the system. For this, farmers will be asked to log into the mobile application, review soil health data, view crop and fertilizer recommendations, and simulate updating their farm records. Meanwhile, DA officials will use the web dashboard to access farmer profiles, view GIS-based maps, and manage crop rules.

A total of 10 participants will be involved in the usability testing, consisting of 7 farmers from Region 10 and 3 agricultural technicians or DA employees. Farmers were selected based on their familiarity with basic smartphone usage and their active involvement in small- to medium-scale farming. The DA personnel were chosen because they represent the system's administrative users responsible for monitoring and validating agricultural data. This mix of participants ensures that both end-user and administrative perspectives are considered during evaluation.

After completing the navigation of the system, the System Usability Scale (SUS) developed by Brooke (1995) will be employed to quantitatively measure user satisfaction. The interactions of the users will be observed, and any difficulties they encounter will be recorded. In addition to the SUS scores, qualitative feedback will be collected through short interviews and observation notes to gain insights into which features are most useful and which aspects require improvement. The evaluation criteria will determine the user satisfaction based on the SUS scores and user feedback. The result of this testing will determine the system's usability rating and this will be a guide for further refinement to ensure that the TANIM system is accessible, user-friendly, and beneficial to the users.

References

Arlyn, T. and Carmelito, L. (2024). Soil fertility assessment and nutrient management for optimizing rice productivity in irrigated lowland areas of northern mindanao. *International Journal of Progressive Research in Engineering, Management and Science*.

Bato, C. (2018). Gis-based suitability mapping of banana in the philippines. *International Society for Southeast Asian Agricultural Sciences (ISSAAS) Philippines*. Online Report.

Calalang, G. D., Bock, L., and Colinet, G. (2014). Establishing a soil reference system for fertility assessment and monitoring at plot level in the highlands of mindanao, philippines. *Eurasian Journal of Soil Science*, 3(3):189–196.

Canatoy, R. (2018). Effects of fertilization on the growth and yield of sweet corn under no tillage in bukidnon, philippines. *International Journal of Scientific and Research Publications (IJSRP)*, 8.

Cicciù, B., Schramm, F., and Schramm, V. B. (2022). Multi-criteria decision making/aid methods for assessing agricultural sustainability: A literature review. *EnvironmentalSciencePolicy*, 138:85–96.

Department of Agriculture Region X (2023). Agricultural production and zoning reports.

Islam, M. R., Oliullah, K., Kabir, M. M., Alom, M., and Mridha, M. F. (2023). Machine learning enabled iot system for soil nutrients monitoring and crop recommendation. *Journal of Agriculture and Food Research*, 14:100880.

Kumar, L. L., Srivani, M., Nishath, M. T., Akhil, T., and Kumar, K. C. (2024a). Monitoring of soil nutrients using soil npk sensor and arduino.

Kumar, R. et al. (2024b). Multi-crop soil health monitoring using iot sensors and machine learning. *Agricultural Systems*.

Lal, R. (2024). Soil degradation effects on human malnutrition and undernutrition. *Medical Research Archives*, 12(10).

Lavanya, R. M., Naidu, S., Laxmi, B., Sukumar, P., and Sucharitha, M. (2024). Optimizing fertilizer recommendations for crops using gradient boosted decision trees with logistic regression and ga. In *2024 Global Conference on Communications and Information Technologies (GCCIT)*, pages 1–6.

Morgan, J. B. and Connolly, E. L. (2013). Plant-soil interactions nutrient uptake. *Nature Education*, 4(2).

Parganiha, V. and Verma, M. (2024). An effective soil analysis and crop yield prediction based on optimised light gbm in smart agriculture. *Wilry online library*, 210.

PhilAtlas. Bukidnon profile. `https://www.philatlas.com/mindanao/r10/bukidnon.html`.

Poliquit, D., Tangaran, L., and Labrador, M. (2019a). Soil fertility evaluation and crop suitability assessment of agricultural soil condition in samar province, philippines. *Journal of the Austrian Society of Agricultural Economics*, 16.

Poliquit, D., Tangaran, L., and Labrador, M. (2019b). Soil fertility evaluation and crop suitability assessment of agricultural soil condition in samar province, philippines. *Journal of the Austrian Society of Agricultural Economics*, 16.

Saaty, T. (2008). Decision making with the analytic hierarchy process. *Int. J. Services Sciences Int. J. Services Sciences*, 1:83–98.

Salvador, E. L. V. (2024). Use of boosting algorithms in household-level poverty measurement: A machine learning approach to predict and classify household wealth quintiles in the philippines. https://arxiv.org/pdf/2407.13061.

Samaniego, A. L. and Gallego, M. (2024). Farm soil testing techniques, soil health, and potential impact of artificial intelligence as factors of small-scale farming soil management. *E3S Web of Conferences*, 595.

Sani, S. R., Ummadi, S. V. S., Thota, S., Muthineni, N., Swargam, V. S. S., and Ravella, T. S. (2023). Crop recommendation system using random forest algorithm in machine learning. *2023 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, pages 501–505.

Sondhiya, R. R. and Singh, V. K. (2024). Intelligent agriculture: Integrating iot and machine learning for soil nutrients and crop selection. *ShodhKosh: Journal of Visual and Performing Arts*, 5.

Sudha, B. and Hooda, B. K. (2022). Decision tree algorithms and their applicability in agriculture for classification. *ResearchGate*.

Temario, A. and Lapoot, C. (2024). Soil fertility assessment and nutrient management for optimizing rice productivity in irrigated lowland areas of northern mindanao. *International Journal of Progressive Research in Engineering, Management and Science*, 4.

Watson, C. J. P., Badayos, R. B., Sanchez, P. B., and Cruz, P. C. S. (2025). Characterization, classification, and crop suitability assessment of selected soil series in ilocos norte, philippines. *Mindanao Journal of Science and Technology*, 23(1):121–150.

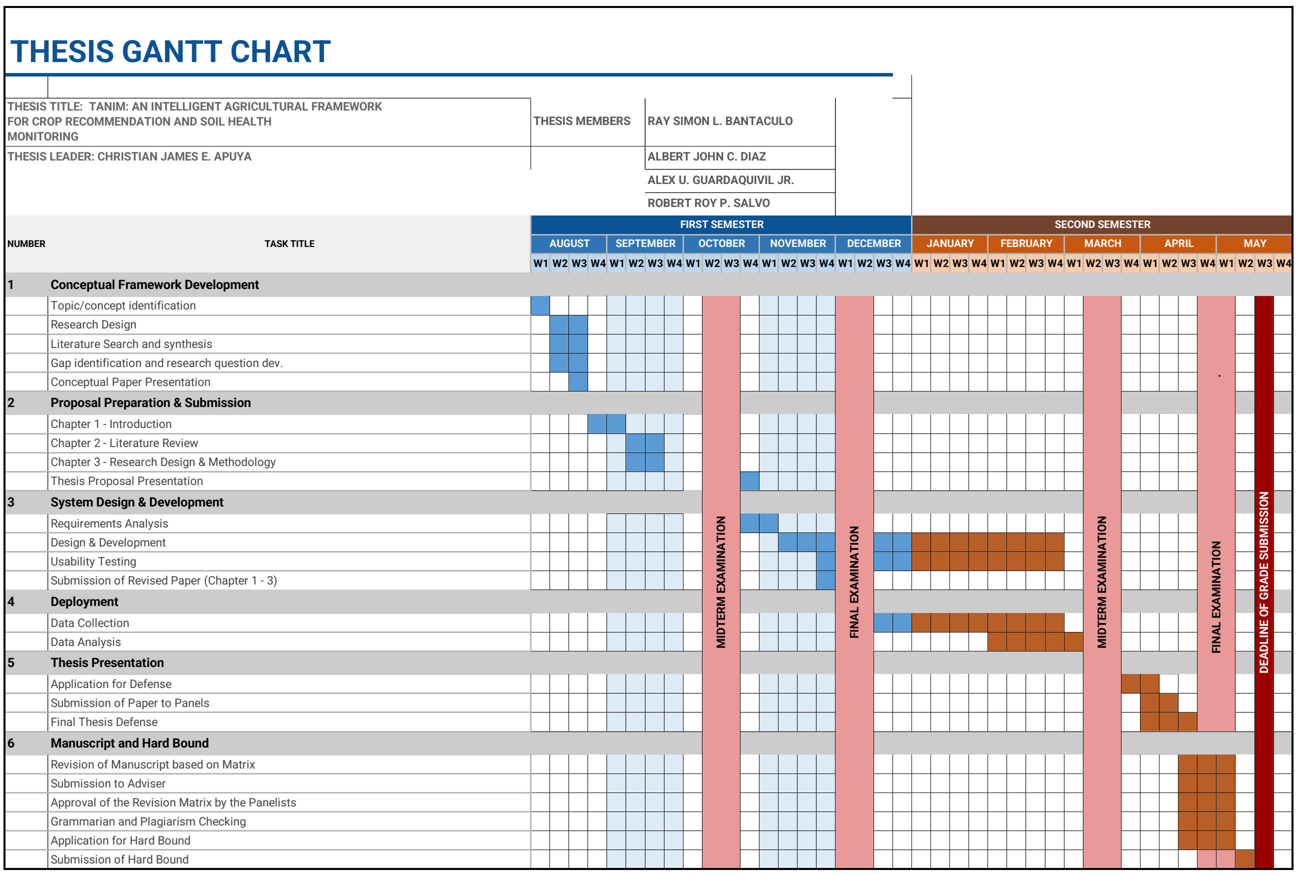
Yin, H., Cao, Y., Marelli, B., Zeng, X., Mason, A., and Cao, C. (2021). Soil sensors and plant wearables for smart and precision agriculture. *Advanced Materials*, 33.

Yuan, Z., Hua, W., He, C., Zhou, J., Zhou, Z., and Xu, F. (2022). Application of multi-criteria decision-making analysis to rural spatial sustainability evaluation: A systematic review. *International Journal of Environmental Research and Public Health*, 19:6572.

APPENDICES

APPENDIX A

GANTT CHART



APPENDIX B

FUNCTIONAL TESTING FORM

A. Hardware and Data Capture

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Sensor powers on and initializes correctly.	Sensor activates and displays readiness.		
2	The sensor captures soil parameters (N, P, K, pH, salinity, temperature, moisture).	Accurate readings displayed on serial monitor or app.		
3	Multiple readings under the same conditions are consistent.	Readings show minimal variance (±5%).		
4	ESP32 collects and processes sensor data properly.	Data packets successfully generated.		
5	The LoRa module transmits data to the gateway.	Data successfully received at gateway.		
6	Data stored correctly in the Firebase database.	Database shows complete and valid entries.		

B. Mobile Application (Farmer Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	User login and authentication.	Users can log in using valid credentials.		
2	Soil health readings display correctly.	Dashboard shows latest sensor readings.		
3	Crop recommendation generation.	App suggests appropriate crops based on soil data.		
4	Fertilizer recommendation with dosage and schedule.	Correct fertilizer type and amount displayed.		
5	Multiple crop recommendations work without error.	The system handles multiple queries smoothly.		
6	Offline mode and data synchronization.	Data syncs automatically once reconnected.		
7	Farm information update.	Farmers can add/edit farm details successfully.		

C. Web Application (DA / Administrator Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Admin login and authentication.	DA officials can securely log in.		
2	Real-time soil health dashboard.	Data from sensors updates in real time.		
3	GIS mapping visualization.	Farm locations shown accurately on the map.		
4	Farmer profile management.	Officials can view and edit farmer data.		
5	Crop rules and recommendation management.	Admins can modify rules and save updates.		
6	Web responsiveness.	The system is accessible and functional on all devices.		

D. System-Wide Features

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	End-to-end data flow.	Data moves seamlessly from sensor $\rightarrow$ gateway $\rightarrow$ Firebase $\rightarrow$ apps.		
2	Error handling for missing/corrupt data.	System displays proper error messages or retries.		
3	Security and access control.	Only authorized users can access restricted features.		
4	Multi-user capability.	Multiple users can operate simultaneously without system crash.		

APPENDIX C

USABILITY TESTING SCALE

System Usability Scale (SUS) Questionnaire						
No.	**Item**	**Frequency**				
		SD	**D**	**N**	**A**	**SA**
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I think the system was easy to use.					
4	I think that I would need the support of a technical person to be able to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found it difficult to navigate the system.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

APPENDIX D

TESTING CONDUCTED

APPENDIX E

LETTER OF CONSENT FOR INTERVIEW

APPENDIX F

LETTER OF REQUEST FOR SOIL TESTING KIT

APPENDIX G

DOCUMENTS REVIEWED

APPENDIX H

INTERVIEW QUESTIONS

APPENDIX I

CALIBRATION OF SOIL SENSOR USING SOIL TESTING KIT

APPENDIX J

DOCUMENTATION OF THE IMPLEMENTATION

APPENDIX K

FUNCTIONAL AND USABILITY TESTING CONDUCTED

APPENDIX L

SOIL LABORATORY RESULTS

APPENDIX M

SENSOR DATA SHEET

APPENDIX N

TURNITIN RESULTS

APPENDIX O

GRAMMARIAN RESULTS